Final Project: 8-bit Pipeline ADC Design

ESE6680 — Mixed Signal Circuit Design and Modeling
University of Pennsylvania

Adithya Selvakumar Krishna Karthikeya Chemudupati

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1 Design Overview

This report presents the design, implementation, and characterization of an 8-bit pipeline analog-to-digital converter (ADC) in the AMI 0.60u C5N (3M, 2P, high-res) process using Cadence. The converter employs a 1.5-bit-per-stage architecture with digital error correction, achieving a sampling rate of 2.5 MHz with the input range of 1–4 V ($V_{\text{cm}} = 2.5\,\text{V}$, $V_{\text{pk}} = 1.5\,\text{V}$).

1.1 Target Specifications

Table 1: ADC target specifications.

Parameter	**Specification**	**Value**
Supply voltage	V_{DD}	5 V
Signal bandwidth	DC to f_{BW}	1 MHz
Sampling frequency	$f_s > 2\,\text{MHz}$	2.5 MHz
Resolution	N	8 bits
Input range	V_{in}	1 V – 4 V
Common-mode voltage	V_{cm}	2.5 V
DNL		≤ 1 LSB
INL		≤ 1 LSB
Optimization goal	FOM	Minimize

1.2 Architecture Selection

The 1.5-bit-per-stage pipeline architecture was selected for its robustness to comparator offsets. Each stage performs a coarse 3-level quantization of its input, subtracts the quantized estimate via a sub-DAC, and amplifies the residue by 2 for the next stage. The redundancy in the 1.5-bit coding allows comparator offsets up to $\pm V_{\text{ref}}/4 = \pm 750\,\text{mV}$ to be corrected digitally, greatly relaxing comparator accuracy requirements.

The full converter consists of:

- Front-end flip-around sample-and-hold amplifier (SHA)
- Six identical 1.5-bit pipeline stages
- One 2-bit flash back-end ADC
- Digital decoder with shift-register timing alignment

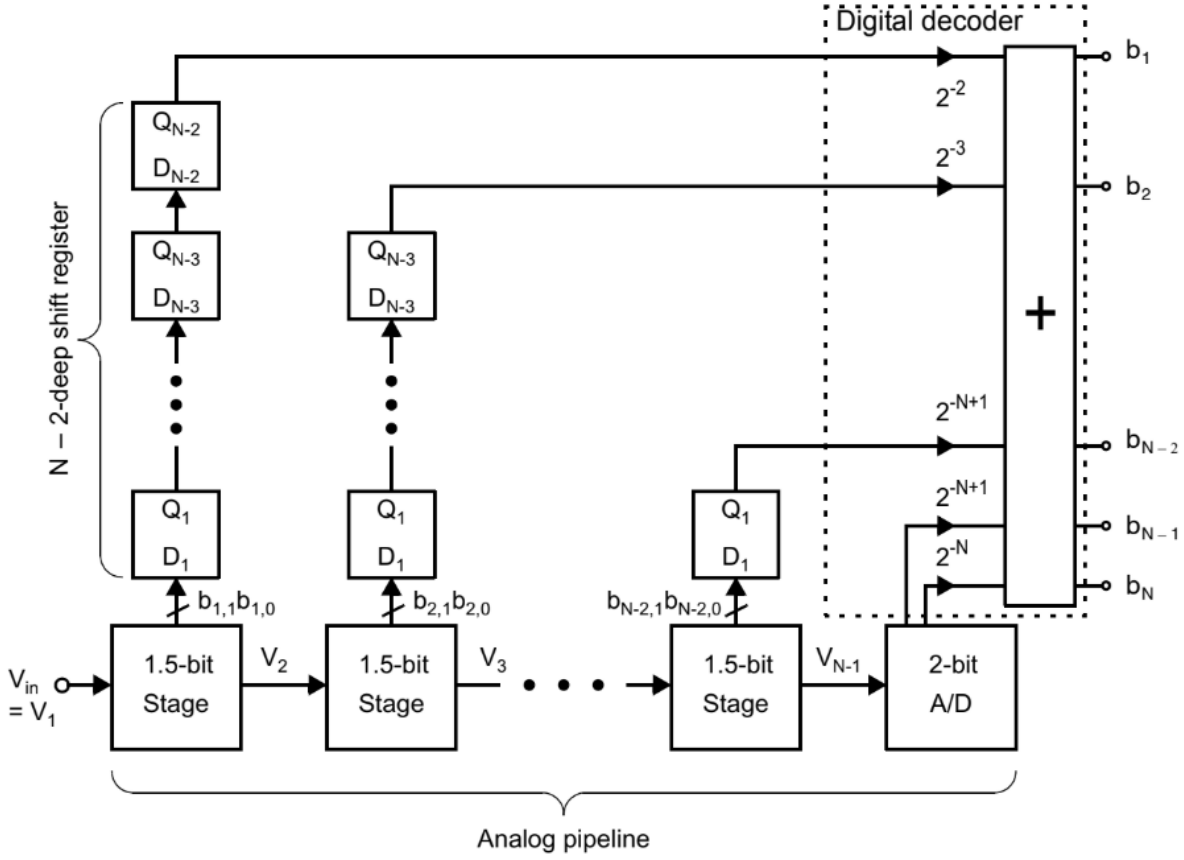


Figure 1: Figure taken from *Analog Integrated Circuit Design* (Second Edition) by Carusone et al. Top-level architecture: SHA → 6× 1.5-bit stages → 2-bit flash → digital decoder.

2 Architecture and Theory of Operation

2.1 1.5-Bit Stage Transfer Function

Each 1.5-bit stage compares its input V_i against two thresholds at $V_{\text{cm}} \pm V_{\text{ref}}/8 = 2.5 \pm 0.375\,\text{V}$ (absolute: 2.125 V and 2.875 V). Based on the comparison result, the stage produces a residue:

$$V_{i+1} = 2V_i - D_i \cdot \frac{V_{\text{ref}}}{4} \tag{1}$$

where $D_i \in \{0, 1, 2\}$ depends on which region V_i falls in:

Table 2: 1.5-bit stage operation.

Region	**Condition**	$b_{i,0}$	$b_{i,1}$	D_i	**Residue**
Low	$V_i < 2.125\,\text{V}$	0	0	0	$2V_i - 1$
Mid	$2.125 < V_i < 2.875\,\text{V}$	0	1	1	$2V_i - 2.5$
High	$V_i > 2.875\,\text{V}$	1	1	2	$2V_i - 4$

The residue output remains within the 1–4 V range for all valid inputs, ensuring every stage in the pipeline sees the same signal range. The redundancy in the 1.5-bit coding allows comparator

offsets up to $\pm V_{\text{ref}}/4 = \pm 750\,\text{mV}$ to be corrected digitally, because an erroneous comparison merely shifts the transition point along the residue plot without pushing the residue out of the valid range for the next stage.

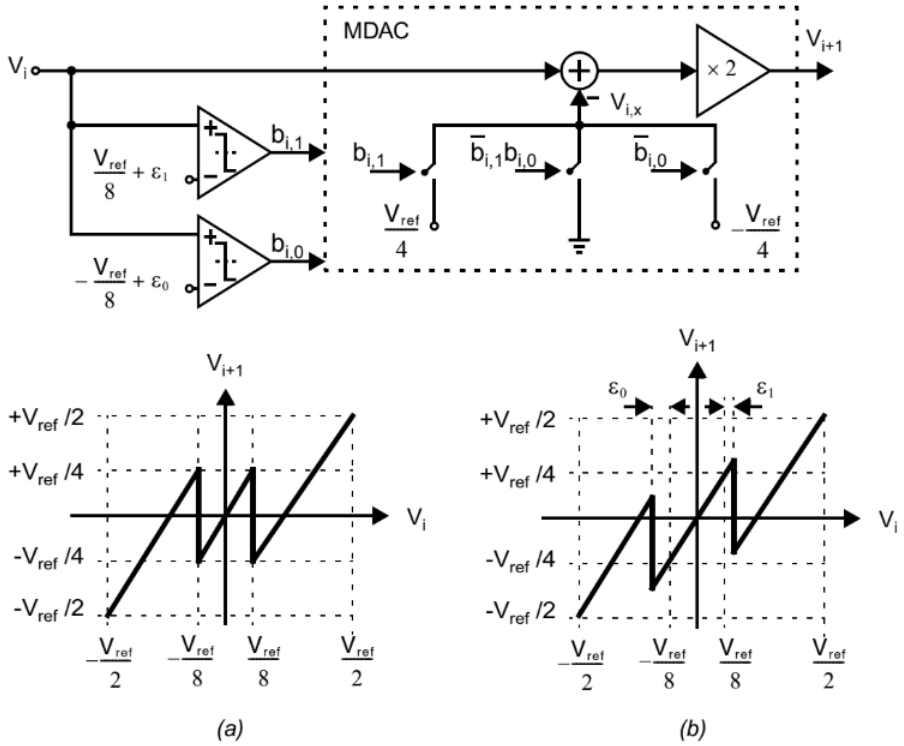


Figure 2: 1.5-bit pipeline stage block diagram and residue transfer characteristic. Figure taken from *Analog Integrated Circuit Design* (Second Edition) by Carusone et al. (a) Ideal transfer function with three segments of slope 2. (b) Transfer function with comparator offsets ε_0, ε_1: the transition points shift but the residue never exceeds the valid range, so subsequent stages still produce correct outputs and the digital decoder recovers the true code.

2.2 MDAC Implementation

The multiplying DAC (MDAC) implements the residue computation $V_{i+1} = 2V_i - D_i \cdot V_{\text{ref}}/4$ using a switched-capacitor circuit. During ϕ_1 (sample phase), the input is sampled onto total capacitance $2C$: a feedback capacitor C and two sub-DAC capacitors of $C/2$ each, all with their top plates connected to the opamp's virtual ground and their bottom plates driven to $V_{\text{cm}} = 2.5\,\text{V}$ through ϕ_{1e}-controlled switches (bottom-plate sampling eliminates signal-dependent charge injection).

During ϕ_2 (amplify phase), the feedback capacitor C flips into the opamp's feedback path, and each $C/2$ sub-DAC capacitor has its bottom plate switched to either $+V_{\text{ref}}/2 = 4\,\text{V}$ or $-V_{\text{ref}}/2 = 1\,\text{V}$ based on the comparator bits $b_{i,0}$ and $b_{i,1}$. A 2:1 analog multiplexer built from CMOS transmission gates selects the appropriate reference for each capacitor. By charge conservation on the top-plate node:

$$V_{\text{out}} = \frac{2C}{C} \cdot V_{\text{in}} - \frac{1}{C}\left(\frac{C}{2} \cdot V_{\text{mux},0} + \frac{C}{2} \cdot V_{\text{mux},1}\right) + V_{\text{cm}} \tag{2}$$

where $V_{\text{mux},0}$ and $V_{\text{mux},1}$ are the reference voltages selected by the sub-DAC muxes. This produces

the three residue segments listed in Table 2. The feedback factor during amplification is $\beta = C/(C+C) = 1/2$.

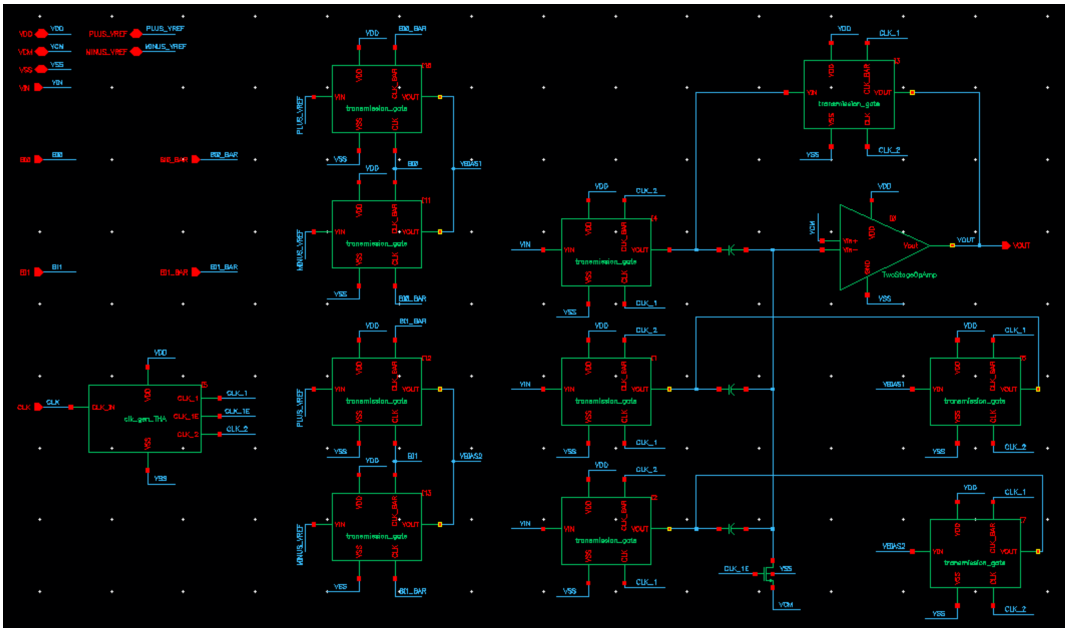


Figure 3: MDAC schematic in Cadence. Left: sub-DAC switching network with transmission gates controlled by BI0/BI1 and their complements, selecting between $+V_{\text{ref}}/2$ (4 V) and $-V_{\text{ref}}/2$ (1 V) for each $C/2$ capacitor. Right: opamp in flip-around configuration with feedback capacitor $C = 2\,\text{pF}$ providing gain of 2.

2.3 Complete 1.5-Bit Stage

The full 1.5-bit stage integrates the sub-ADC (two StrongARM comparators with NAND SR latches) and the MDAC into a single cell. Each comparator receives V_{in} on its positive input and a reference threshold on its negative input ($V_{\text{ref+}} = 2.875\,\text{V}$ or $V_{\text{ref-}} = 2.125\,\text{V}$). The StrongARM evaluates when ϕ_2 is asserted (CLK high) and produces rail-to-rail complementary outputs, but *resets both outputs to V_{DD}* during ϕ_1 (precharge phase). A NAND SR latch after each comparator holds the decision through the precharge: since the NAND SR latch has active-low inputs, both inputs at V_{DD} naturally produce the HOLD condition, requiring no clock gating.

A key polarity detail: the StrongARM outputs are active-low (the "winning" output pulls low during evaluation), so after the SR latch, the `Q_BAR` output—not `Q`—carries the correct active-high bit for the MDAC. Routing `Q_BAR` to $b_{i,0}$ and $b_{i,1}$ produces the correct sub-DAC switching without additional logic.

Each stage also contains an internal non-overlapping clock generator that accepts a single master clock input (CLK) and produces ϕ_1, ϕ_{1e} (leading ϕ_1 by $\sim$1 ns), and ϕ_2.

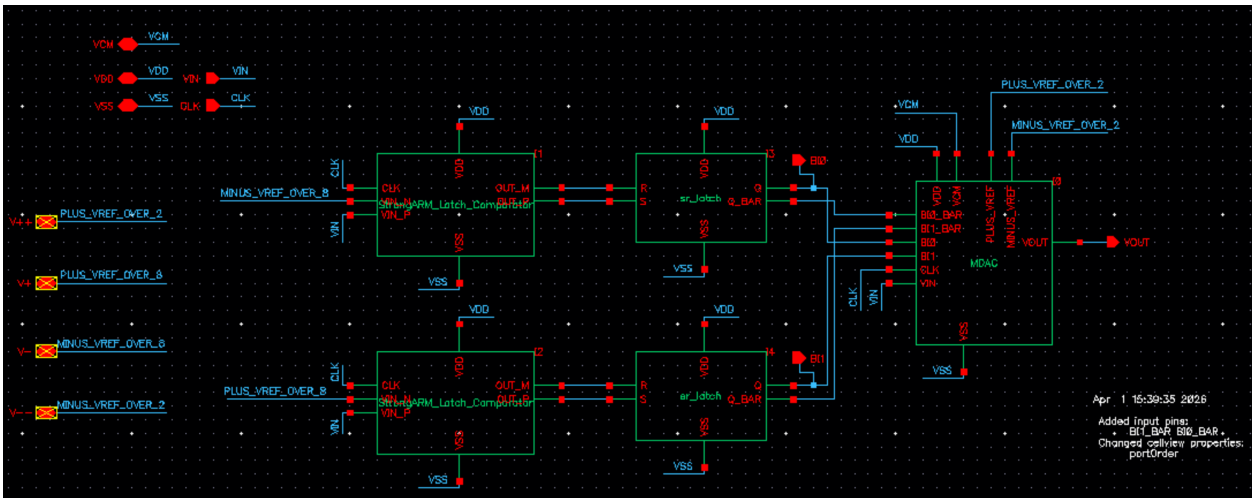


Figure 4: Complete 1.5-bit pipeline stage schematic in Cadence. Two StrongARM comparators (left) compare V_{in} against $V_{\text{ref+}} = 2.875\,\text{V}$ and $V_{\text{ref-}} = 2.125\,\text{V}$, each followed by a NAND SR latch. The latch `Q_BAR` outputs drive the MDAC sub-DAC muxes (right), which generate the residue V_{i+1}.

2.4 Opamp Requirements for the MDAC

The first-stage MDAC has the most stringent requirements since its errors are not attenuated by subsequent gain stages. With $f_s = 2.5\,\text{MHz}$ ($T_s = 400\,\text{ns}$, each phase $\approx 200\,\text{ns}$) and 8-bit accuracy requiring $\sim$9 time constants of settling:

$$\tau_{\text{CL,max}} = \frac{200\,\text{ns}}{9} = 22.2\,\text{ns} \tag{3}$$

$$f_{\text{CL,min}} = \frac{1}{2\pi \cdot 22.2\,\text{ns}} = 7.2\,\text{MHz} \tag{4}$$

$$f_{u,\text{min}} = \frac{f_{\text{CL,min}}}{\beta} = \frac{7.2}{0.5} = 14.4\,\text{MHz} \tag{5}$$

DC gain must exceed $1/(\beta \cdot 2^{-9}) \approx 1024$, or 60 dB, to keep gain error below $\frac{1}{2}$ LSB. Slew rate: the worst-case output step is $V_{\text{ref}}/2 = 1.5\,\text{V}$, giving $\text{SR}_{\text{min}} = 1.5\,\text{V}/80\,\text{ns} = 18.75\,\text{V}/\mu\text{s}$ (allocating 40% of the settling window to slewing).

The Project 1 opamp comfortably exceeds all requirements: 84.2 dB gain (vs. 60 dB), 27.2 MHz GBW (vs. 14.4 MHz), 70.9° phase margin (vs. 60°), and $25.3\,\text{V}/\mu\text{s}$ slew rate (vs. $18.75\,\text{V}/\mu\text{s}$).

2.5 2-Bit Flash Back-End ADC

The final pipeline stage is a 2-bit flash ADC that resolves the residue from Stage 6 into two bits without residue amplification. Three StrongARM comparators quantize the input into four levels, with thresholds placed at quarter-points of the input range: $V_{\text{cm}} - V_{\text{ref}}/4 = 1.75\,\text{V}$, $V_{\text{cm}} = 2.5\,\text{V}$, and $V_{\text{cm}} + V_{\text{ref}}/4 = 3.25\,\text{V}$. Each comparator output feeds a NAND SR latch, identical to the 1.5-bit stage design, to hold the decision through the precharge phase.

The three thermometer-coded comparator outputs C_1, C_2, C_3 (where $C_k = 1$ when V_{in} exceeds threshold k) are converted to a 2-bit binary code using two NAND gates:

$$b_{\text{MSB}} = C_2 \quad \text{(direct wire from Q of middle SR latch)}$$

$$b_{\text{LSB}} = \text{NAND}\big(\text{NAND}(C_1, \overline{C}_2),\ \overline{C}_3\big)$$

This logic implements $b_{\mathrm{LSB}} = C_1 \cdot \overline{C}_2 + C_3$, which is high in the second and fourth quantization regions. The $\overline{C}_2$ and $\overline{C}_3$ signals are taken directly from the `Q_BAR` outputs of the corresponding SR latches, requiring no additional inverters. The first NAND (I7) computes NAND$(C_1, \overline{C}_2)$, and the second NAND (I6) combines this with $\overline{C}_3$ to produce b_{LSB}.

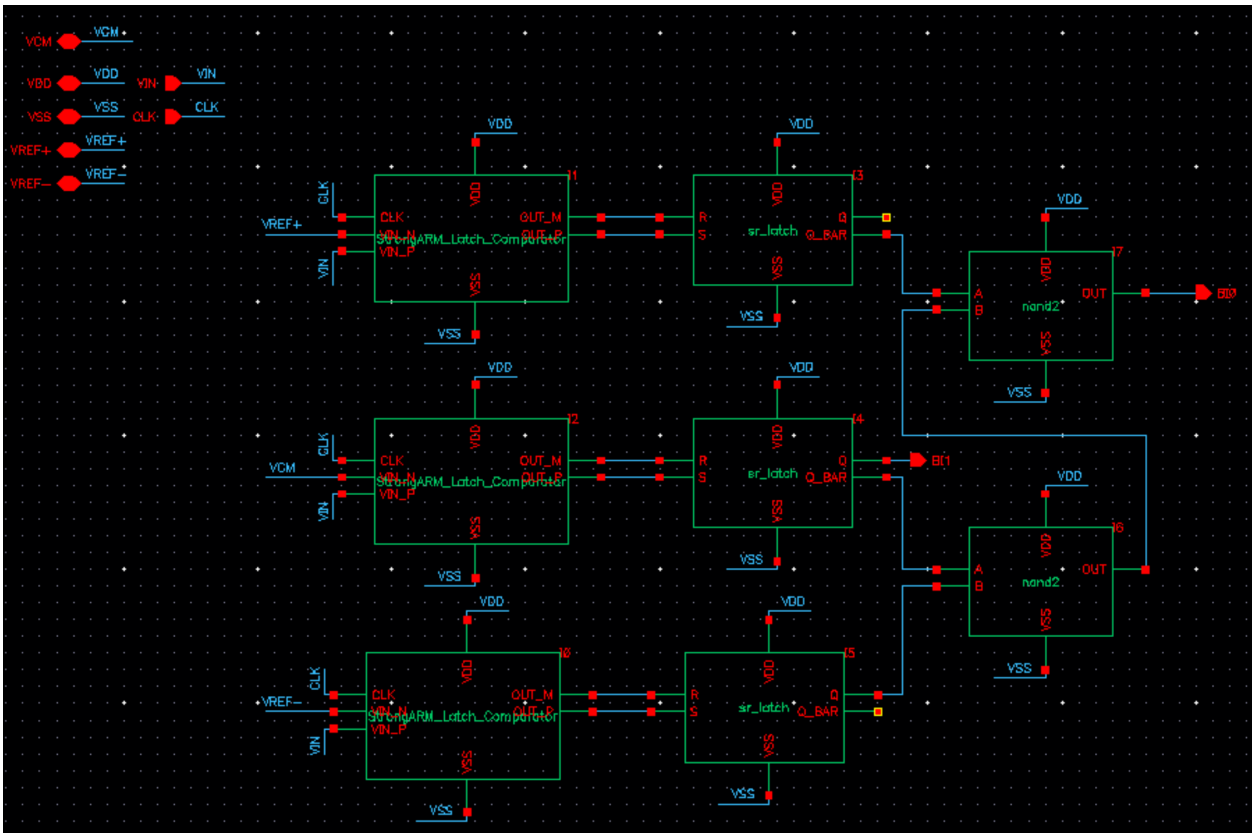


Figure 5: 2-bit flash back-end ADC schematic in Cadence. Three StrongARM comparators (I1, I2, I16) compare V_{in} against VREF+ (3.25 V), VCM (2.5 V), and VREF− (1.75 V), each followed by an SR latch. The middle SR latch Q output provides b_{MSB} (BI1) directly. Two NAND gates (I7, I6) implement the thermometer-to-binary encoder for b_{LSB} (BI0).

2.6 Digital Decoder

The six 1.5-bit stages and the 2-bit flash back-end together produce 14 raw bits (two per stage). The digital decoder must align these bits in time—since each stage has one half-period of latency relative to the previous—and then combine them via a weighted addition with carry propagation to produce the final 8-bit output code.

2.6.1 Decoder Arithmetic

Each 1.5-bit stage produces a digit $d_i = b_{i,0} + b_{i,1} \in \{0, 1, 2\}$, and the 2-bit flash produces $d_7 = 2\, b_{7,\mathrm{MSB}} + b_{7,\mathrm{LSB}} \in \{0, 1, 2, 3\}$. The reconstructed input code is:

$$\mathrm{code} = d_1 \times 64 + d_2 \times 32 + d_3 \times 16 + d_4 \times 8 + d_5 \times 4 + d_6 \times 2 + d_7 \tag{6}$$

Since each d_i can exceed 1, carries propagate between adjacent digit positions. Each stage's bit pair is first decomposed by a half adder into a sum $s_i = b_{i,0} \oplus b_{i,1}$ and a carry $c_i = b_{i,0} \cdot b_{i,1}$, where s_i

contributes to the current bit position and c_i propagates to the next higher position. A ripple-carry adder chain then resolves all inter-stage carries:

$$\begin{aligned}
\text{OUT}[0] &= b_{7,\text{LSB}} \\
\text{OUT}[1],\ \text{carry}_1 &= \text{half_add}(s_6,\ b_{7,\text{MSB}}) \\
\text{OUT}[k],\ \text{carry}_k &= \text{full_add}(s_{7-k},\ c_{7-k},\ \text{carry}_{k-1}) \quad \text{for } k = 2, \ldots, 6 \\
\text{OUT}[7] &= \text{half_add}(c_1,\ \text{carry}_6) \quad \text{(sum only)}
\end{aligned}$$

The implementation requires 8 half adders (one per stage to decompose d_i, plus one for the OUT[1] and OUT[7] positions), 5 full adders (for OUT[2] through OUT[6]), and 2 minimum-size inverters to buffer $b_{7,\text{LSB}}$ to OUT[0].

2.6.2 Shift-Register Timing Alignment

Because the pipeline stages operate sequentially, each stage's bit pair becomes valid one half-period later than the previous stage's. Stage 1's bits are the earliest; the flash's bits are the latest. Before the combinational adder can operate, all 14 bits corresponding to the *same input sample* must arrive simultaneously.

This alignment is achieved with shift registers of decreasing depth built from master-slave D flip-flops. Each DFF, clocked on alternating phases CLK_A and CLK_B, provides exactly one half-period of delay. Stage 1's bits pass through 6 DFFs (6 half-periods of delay), Stage 2's through 5, and so on down to Stage 6 which needs only 1 DFF. The flash bits require no delay and enter the decoder directly.

The first DFF in each chain is clocked on the *opposite* phase from when that stage's bits become valid, so it cleanly captures the settled data. Subsequent DFFs alternate between CLK_A and CLK_B:

Stage 1 (6 DFFs): $\text{CLK}_A \to \text{CLK}_B \to \text{CLK}_A \to \text{CLK}_B \to \text{CLK}_A \to \text{CLK}_B$

Stage 2 (5 DFFs): $\text{CLK}_B \to \text{CLK}_A \to \text{CLK}_B \to \text{CLK}_A \to \text{CLK}_B$

Stage 3 (4 DFFs): $\text{CLK}_A \to \text{CLK}_B \to \text{CLK}_A \to \text{CLK}_B$

Stage 4 (3 DFFs): $\text{CLK}_B \to \text{CLK}_A \to \text{CLK}_B$

Stage 5 (2 DFFs): $\text{CLK}_A \to \text{CLK}_B$

Stage 6 (1 DFF): CLK_B

Flash (0 DFFs): direct

Each stage has two bits ($b_{i,0}$ and $b_{i,1}$), so two identical shift-register chains run in parallel per stage. The total is $2 \times (6 + 5 + 4 + 3 + 2 + 1) = 42$ DFFs.

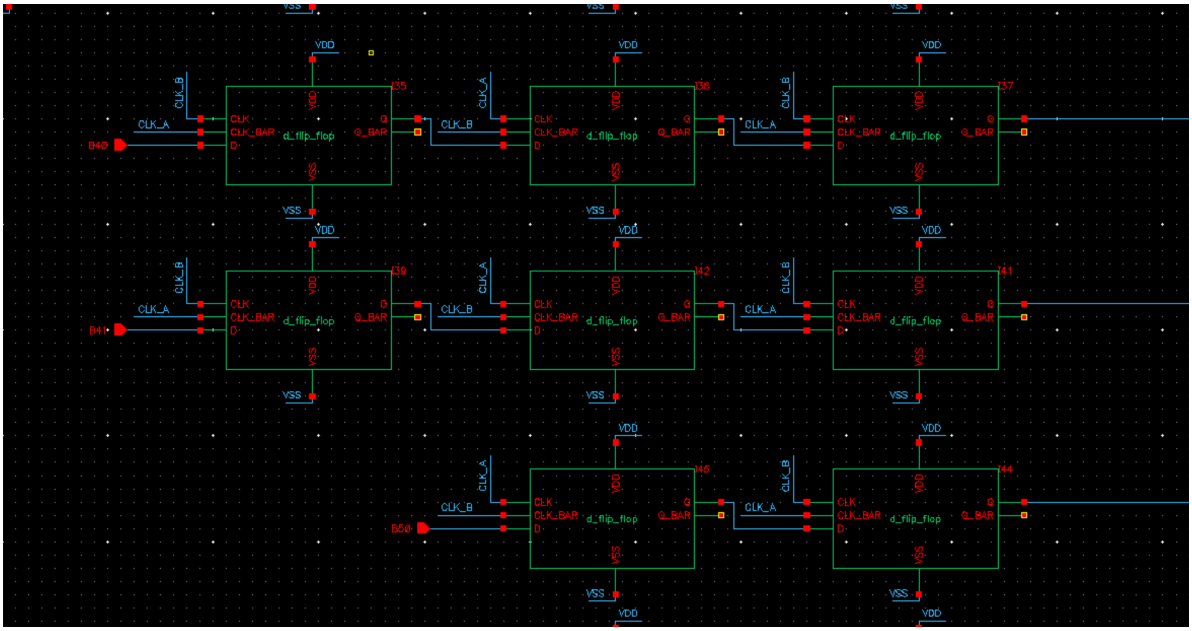


Figure 6: Close-up of DFF shift-register chains inside `adc_digital_logic`, showing Stages 4 and 5. Stage 4 bits (B40, B41) each pass through 3 DFFs with alternating $\text{CLK}_A/\text{CLK}_B$ clocking. Stage 5 bit B50 passes through 2 DFFs. The CLK and CLK_BAR pins of each DFF connect to the appropriate phase, providing one half-period of delay per flip-flop.

2.6.3 Combinational Decoder Implementation

The combinational portion of the decoder (`digital_decoder` cell) takes in the time-aligned 14 bits and produces 8 output bits using the adder chain described above. The schematic is split across two figures for clarity.

Figure 7 shows the upper portion: half adders for Stages 1–3 decompose each bit pair ($b_{i,0}$, $b_{i,1}$) into sum s_i and carry c_i, and full adders compute OUT[4] through OUT[6] with carry propagation. At the top, a final half adder computes OUT[7] from Stage 1's carry c_1 and the ripple carry carry_6.

Figure 8 shows the lower portion: half adders for Stages 4–6, full adders for OUT[2] and OUT[3], and the bottom-level half adder that combines Stage 6's sum s_6 with $b_{7,\text{MSB}}$ to produce OUT[1] and carry_1. The flash LSB ($b_{7,\text{LSB}}$) passes through two cascaded minimum-size inverters (I14, I15) as a buffer to produce OUT[0].

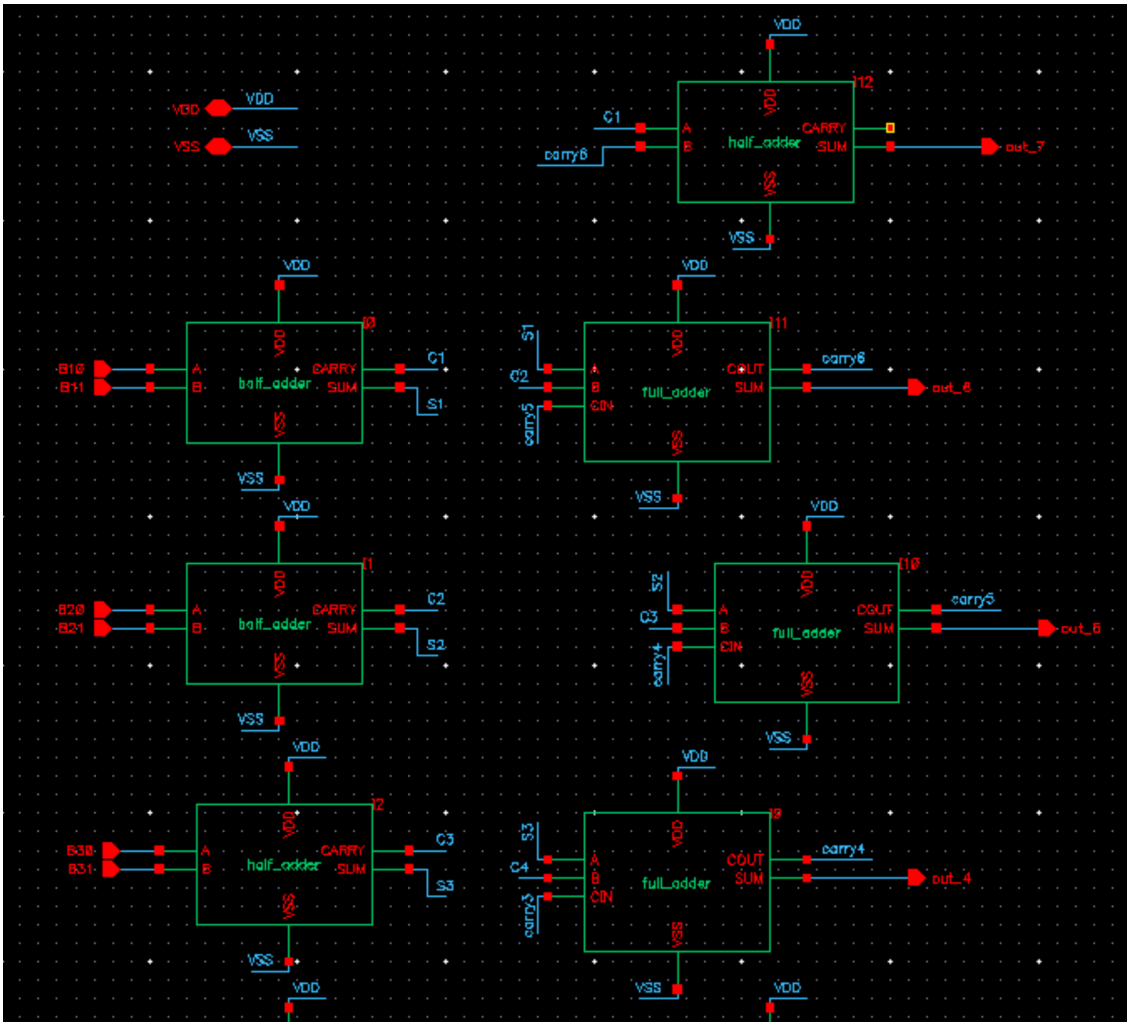


Figure 7: Combinational decoder upper portion (`digital_decoder` cell). Half adders decompose Stages 1–3 bit pairs into sums and carries. Full adders with ripple carry produce OUT[4]–OUT[6]. The topmost half adder computes OUT[7] $= c_1 \oplus \text{carry}_6$.

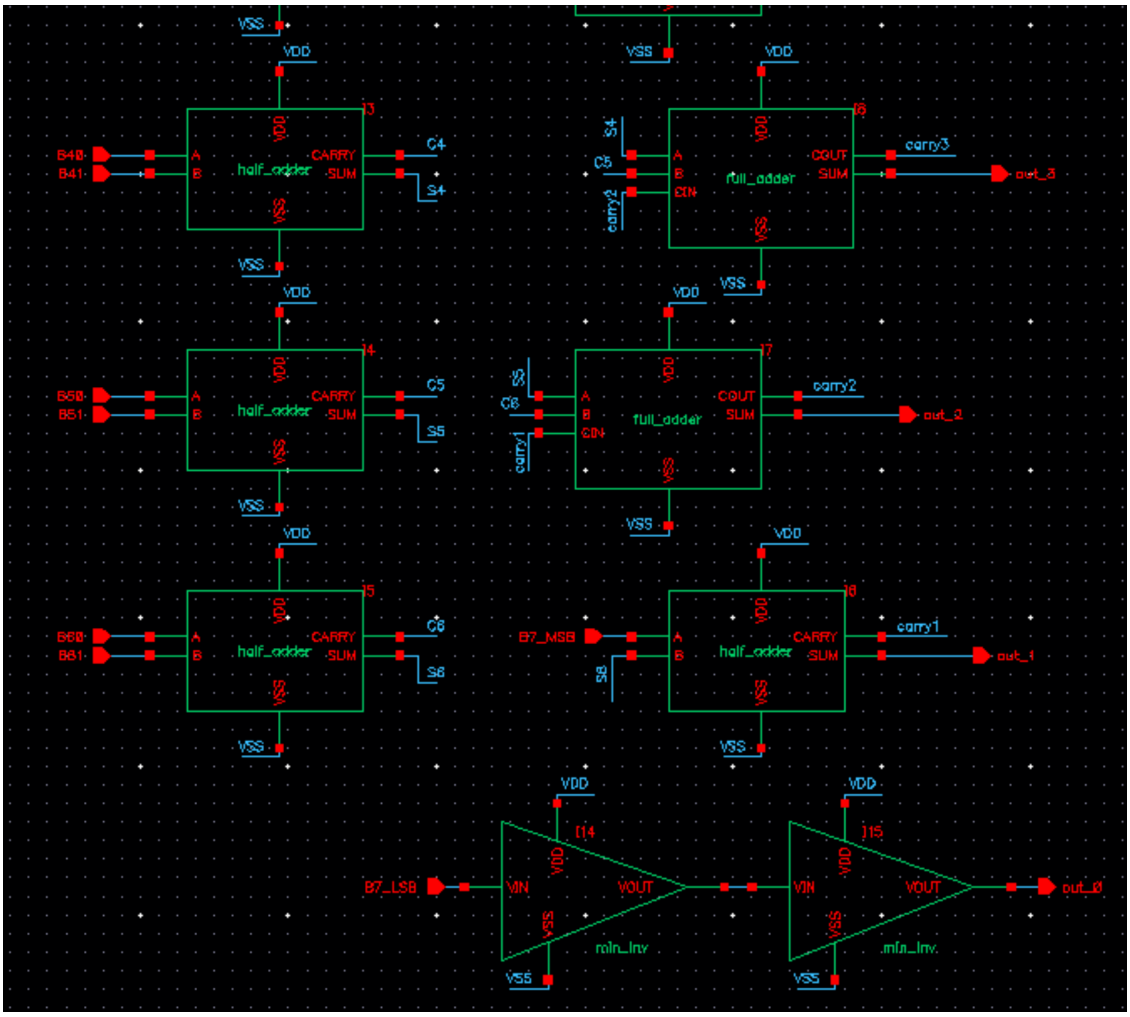


Figure 8: Combinational decoder lower portion (`digital_decoder` cell). Half adders decompose Stages 4–6 bit pairs. Full adders produce OUT[2] and OUT[3]. A half adder combines s_6 and $b_{7,\mathrm{MSB}}$ to produce OUT[1]. Two minimum-size inverters buffer $b_{7,\mathrm{LSB}}$ to OUT[0].

2.6.4 Complete Digital Logic Block

The complete digital logic block (`adc_digital_logic` cell) encapsulates both the 42-DFF shift-register timing alignment network and the `digital_decoder` combinational logic into a single hierarchical cell. External inputs are the 14 raw bits from the six 1.5-bit stages and the flash (B10, B11, B20, B21, . . . , B61, B7_LSB, B7_MSB), plus CLK_A, CLK_B, VDD, and VSS. The eight outputs (out_0 through out_7) are the final reconstructed 8-bit code.

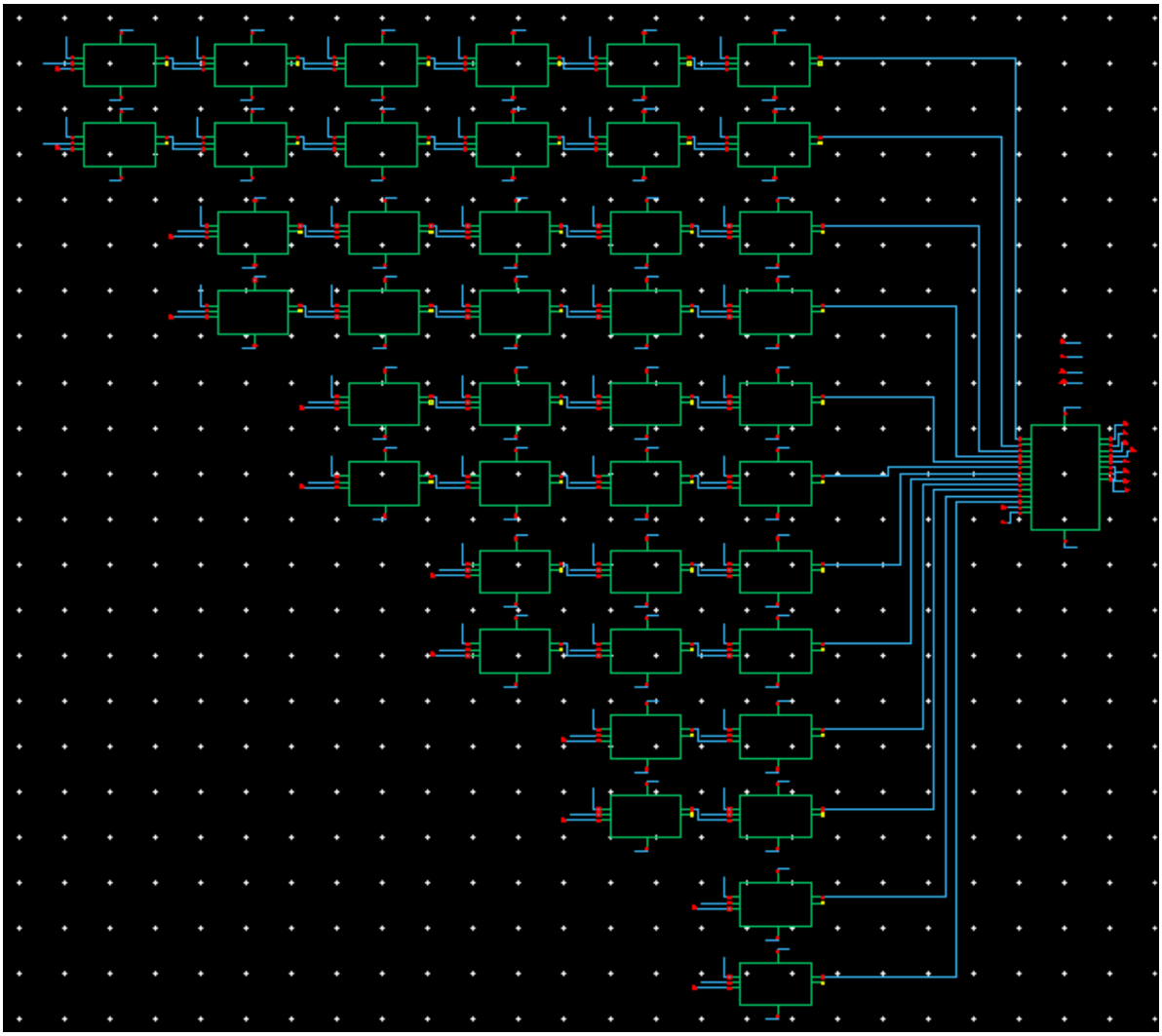


Figure 9: Complete `adc_digital_logic` cell (zoomed out). The 42 DFFs form a characteristic triangular arrangement: Stage 1's chains (top two rows, 6 DFFs each) are the longest, decreasing to Stage 6 (1 DFF each) near the bottom. All shift-register outputs feed into the `digital_decoder` block on the right, which produces OUT[0]–OUT[7].

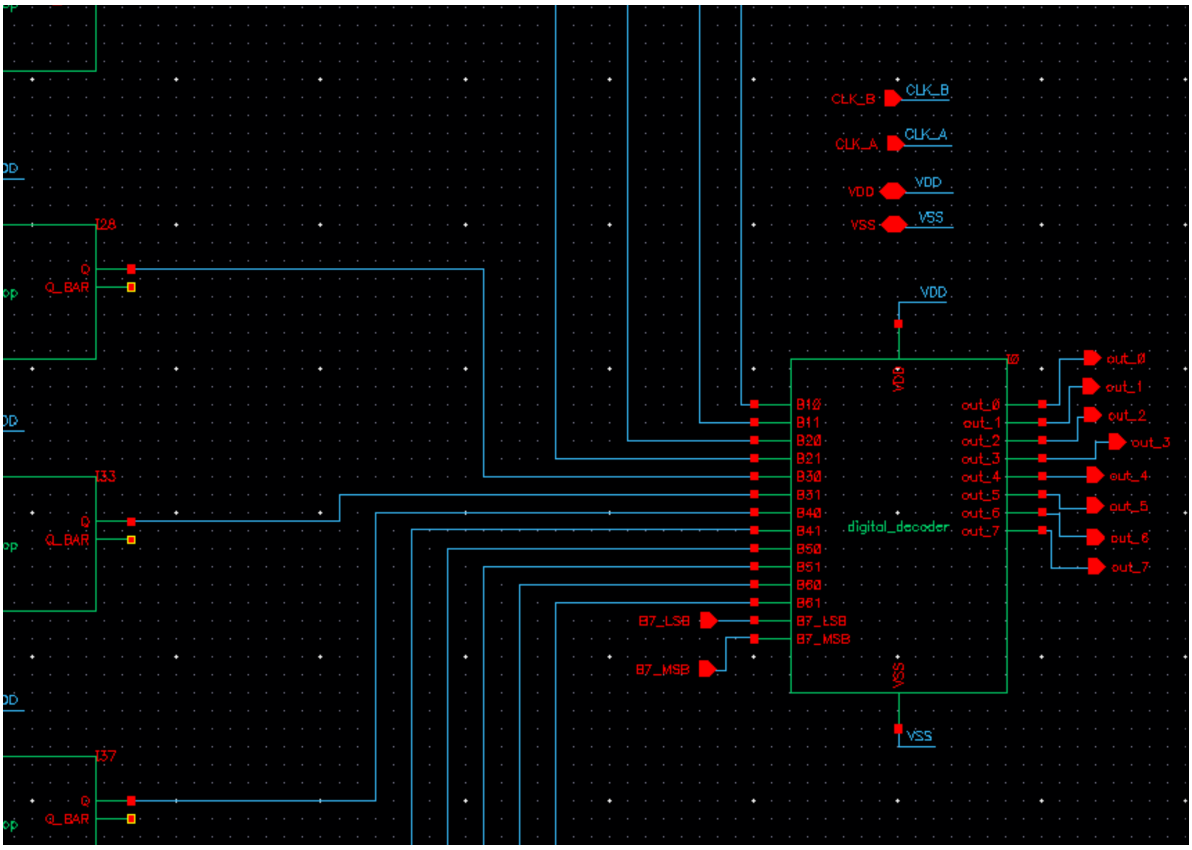


Figure 10: Instantiation of the `digital_decoder` block within `adc_digital_logic`. The 14 time-aligned bit inputs (B10–B61, B7_LSB, B7_MSB) enter on the left. CLK_A and CLK_B are supplied for the shift registers. The eight decoded outputs (out_0–out_7) exit on the right to the top-level ADC pins.

3 Component Design

3.1 Two-Stage Miller-Compensated Opamp

The opamp is the two-stage Miller-compensated design from Project 1, using a PMOS differential input pair (M1/M2) with NMOS active loads (M3/M4), a PMOS common-source second stage (M7) with NMOS current sink (M6), and a series R_C–C_C compensation network for RHP zero cancellation.

Table 3: Opamp transistor dimensions (from Project 1).

Transistor	Reference	Type	W (μm)	L (μm)
P4	M8 (ref.)	PMOS	40.05	2.4
P3	M5 (tail)	PMOS	40.05	2.4
P0	M1 (input)	PMOS	49.95	1.8
P1	M2 (input)	PMOS	49.95	1.8
P2	M7 (2nd)	PMOS	229.95	2.4
N1	M3 (load)	NMOS	8.7	1.8
N3	M4 (load)	NMOS	8.7	1.8
N2	M6 (sink)	NMOS	67.35	1.8

Compensation: $C_C = 750\,\text{fF}$, $R_C = 6\,\text{k}\Omega$, $I_{\text{ref}} = 20\,\mu\text{A}$.

Table 4: Opamp measured performance (from Project 1 characterization).

Parameter	Measured	ADC Requirement
DC Gain	84.2 dB	≥ 54 dB
GBW	27.2 MHz	≥ 14.4 MHz
Phase Margin	70.9°	$\geq 60°$
Slew Rate	25.3 V/μs	≥ 18.75 V/μs
Power	0.85 mW	—

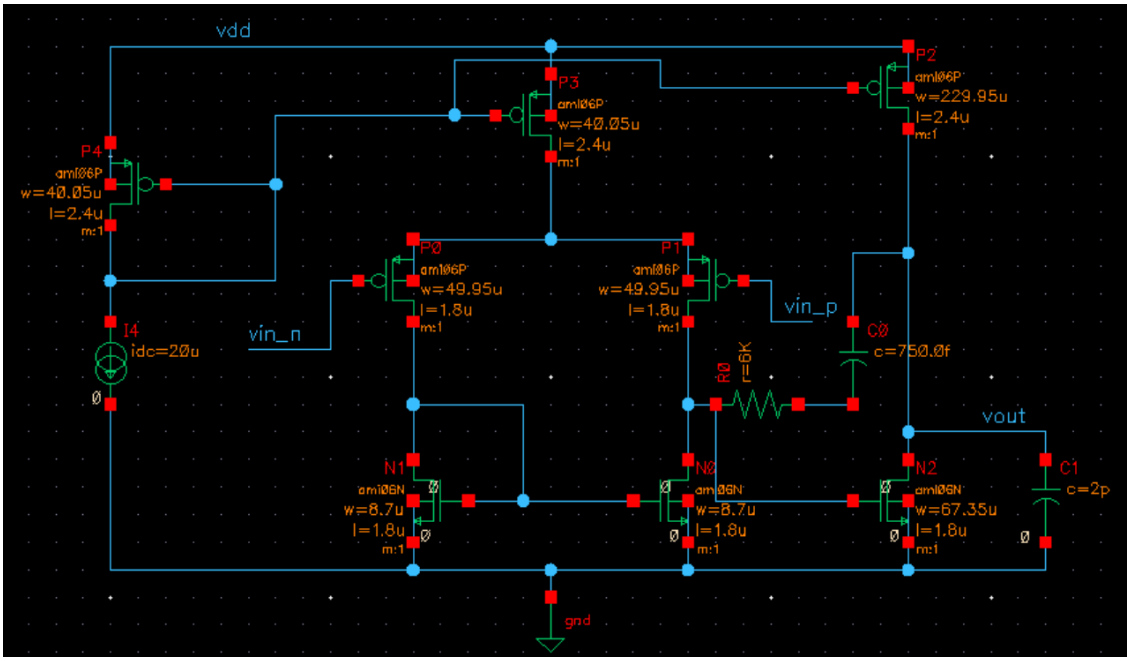


Figure 11: Two-stage Miller-compensated opamp (Project 1 design).

3.2 StrongARM Dynamic Latch Comparator

The sub-ADC comparator is a StrongARM dynamic latch (from Project 2, Problem 3). All transistors are minimum-sized ($W/L = 1.5/0.6\,\mu$m). The comparator evaluates when the clock goes high (NMOS tail turns on, PMOS precharges when clock is low) and produces complementary rail-to-rail outputs.

Key characteristics at $V_{\text{cm}} = 2.5\,\text{V}$, $V_{\text{DD}} = 5\,\text{V}$:

- Time-to-decision: 1.15 ns for $\Delta V_{\text{in}} = 50\,\text{mV}$
- Offset with intentional mismatch: 95.2 mV (tolerance: $\pm 750\,\text{mV}$)
- Zero static power consumption (dynamic only)

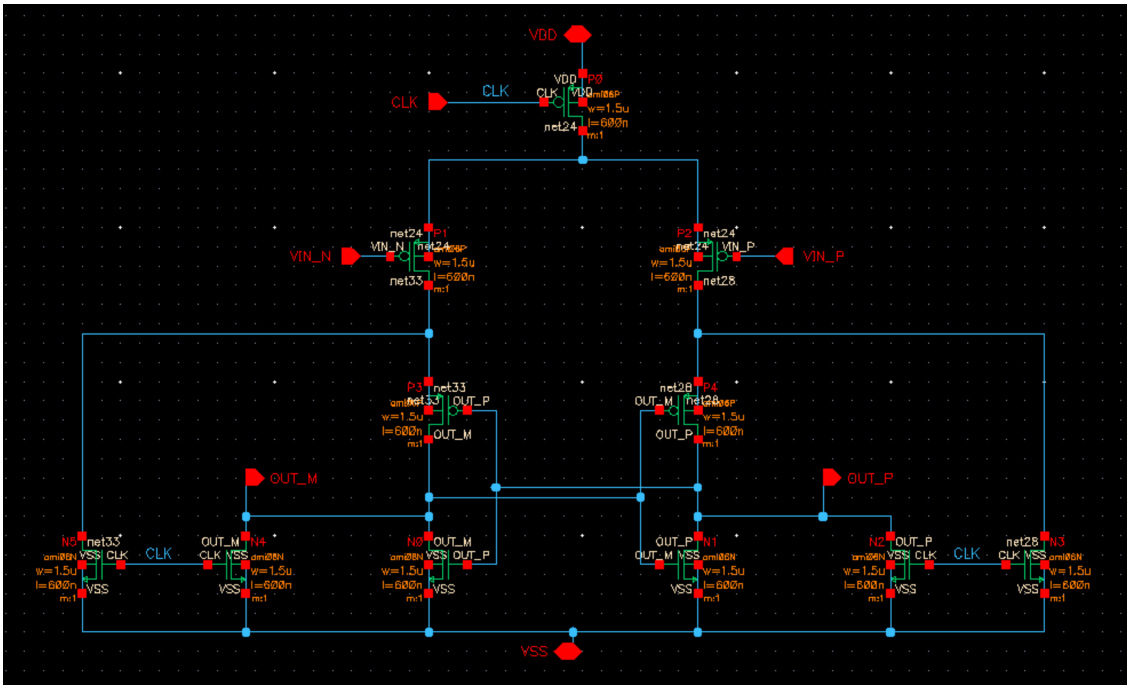


Figure 12: StrongARM dynamic latch comparator.

3.3 NAND SR Latch

The StrongARM outputs precharge to V_{DD} during the reset phase (ϕ_2), requiring a storage element to hold the comparator decision. A NAND-based SR latch (two cross-coupled NAND gates) serves this purpose. With active-low inputs, both inputs at V_{DD} (the precharge state) produce the HOLD condition, preserving the previous decision without clock gating.

The SR latch Q_BAR output is routed to the MDAC bit inputs (not Q), because the StrongARM outputs are active-low: the winning output goes low during evaluation.

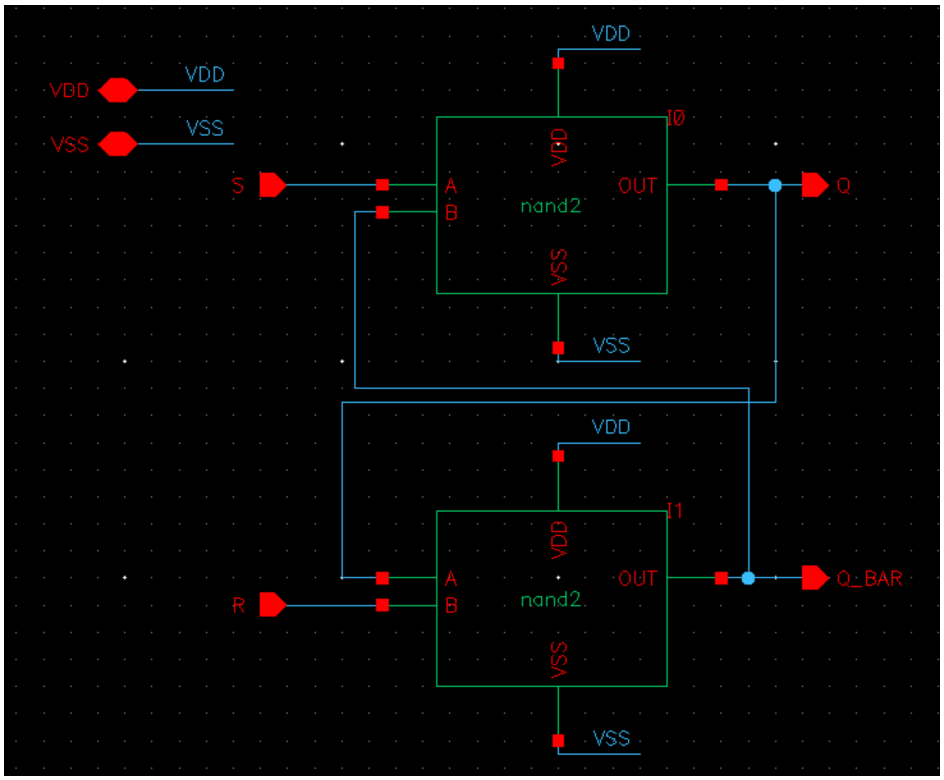


Figure 13: NAND-based SR latch for comparator output storage.

3.4 Non-Overlapping Clock Generator

A master clock generator at the top level of the ADC takes the external CLK input and produces two non-overlapping phases: CLK_A and CLK_B. No early phases are generated at this level.

The SHA and even-numbered 1.5-bit stages (2, 4, 6) receive CLK_A; odd-numbered 1.5-bit stages (1, 3, 5) receive CLK_B. Each MDAC (and the SHA) contains its own internal clock generator that takes its assigned master clock and locally produces three phases: ϕ_1, ϕ_{1e} (leading ϕ_1 by $\sim$1 ns for bottom-plate sampling), and ϕ_2 (non-overlapping with ϕ_1). Generating the early phase locally inside each MDAC eliminates the need to route a separate ϕ_{1e} signal across the full chip.

3.5 Flip-Around Sample-and-Hold Amplifier

The front-end SHA (from Project 2, Problem 1) uses the flip-around topology (Figure 14). During ϕ_1, a CMOS transmission gate samples V_{in} onto a 2 pF capacitor while the opamp is held in reset via a ϕ_1 feedback switch. A ϕ_{1e}-controlled bottom-plate switch (minimum-size NMOS, $W/L = 1.5/0.6\,\mu$m) opens slightly before the top-plate sampling switch to implement bottom-plate sampling, ensuring that the signal-dependent charge injection from the top-plate switch does not corrupt the sampled voltage.

During ϕ_2, the sampling capacitor flips into the feedback path, and charge conservation forces $V_{\text{out}} = V_{\text{in}}$. The feedback factor $\beta = C/(C + C_p) \approx 1$ provides maximum closed-loop bandwidth. The SHA ensures that Stage 1's comparators and MDAC both see the same held input value, preventing errors from input changes during the Stage 1 evaluation window.

Table 5: Flip-around SHA component sizing (from Project 2).

Component	**Type**	**Size**
Sampling capacitor C	MIM cap	2 pF
ϕ_1 sampling switch	CMOS TG	NMOS 3/0.6, PMOS 6/0.6
ϕ_{1e} bottom-plate switch	NMOS	1.5/0.6 (min)
ϕ_1 reset switch	CMOS TG	NMOS 3/0.6, PMOS 6/0.6
ϕ_2 feedback switch	CMOS TG	NMOS 3/0.6, PMOS 6/0.6
Opamp	Two-stage Miller	Table 3

The 2:1 PMOS-to-NMOS width ratio in the transmission gates equalizes R_{on} across the 1–4 V input swing. At $V_{\text{in}} = 4\,\text{V}$, body effect pushes V_{tn} to $\sim$1.66 V and the NMOS cuts off entirely; the PMOS ($|V_{\text{ov}}| = 3.15\,\text{V}$) carries the full current. At $V_{\text{in}} = 1\,\text{V}$, the PMOS cuts off ($|V_{\text{ov}}| = -0.15\,\text{V}$) and the NMOS ($V_{\text{ov}} = 2.84\,\text{V}$) takes over. Worst-case parallel $R_{\text{on}} \approx 1.3\,\text{k}\Omega$ occurs near $V_{\text{in}} = 3.5\,\text{V}$, giving a sampling time constant $\tau_{\text{acq}} = R_{\text{on}} \cdot C = 1.3\,\text{k}\Omega \times 2\,\text{pF} = 2.6\,\text{ns}$. With 200 ns available in the sample phase, this provides $\sim$77 time constants—more than sufficient for signal acquisition.

The bottom-plate switch charge injection pedestal is signal-independent: at turnoff, both terminals sit at $V_{\text{cm}} = 2.5\,\text{V}$, injecting $\Delta V \approx 0.6\,\text{mV}$—a DC offset easily tolerable for 8-bit accuracy ($1\,\text{LSB} = 11.7\,\text{mV}$).

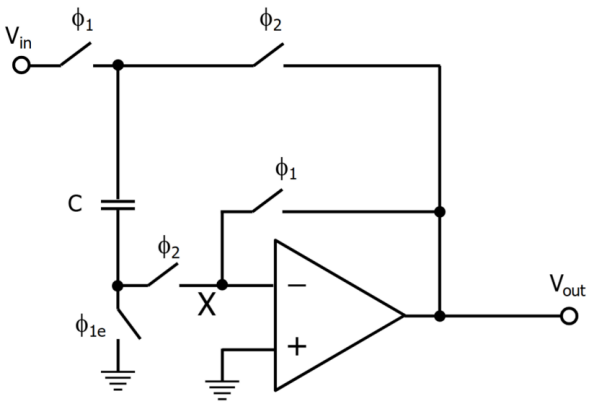


Figure 14: Flip-around sample-and-hold amplifier topology. Figure taken from Lecture 10, ESE6680, by Prof. Tania Khanna.

3.6 Reference Voltage Ladder

A single resistor ladder from V_{DD} (5 V) to ground generates all six reference voltages needed by the pipeline stages and flash back-end. Seven resistors produce six taps:

Table 6: Reference ladder design.

Tap Voltage	Purpose	Resistor Above
4.000 V	$+V_{\text{ref}}/2$ (MDAC)	16 kΩ
3.250 V	$+V_{\text{ref}}/4$ (flash)	12 kΩ
2.875 V	$+V_{\text{ref}}/8$ (1.5-bit)	6 kΩ
2.125 V	$-V_{\text{ref}}/8$ (1.5-bit)	12 kΩ
1.750 V	$-V_{\text{ref}}/4$ (flash)	6 kΩ
1.000 V	$-V_{\text{ref}}/2$ (MDAC)	12 kΩ

Total ladder resistance is 80 kΩ, drawing 62.5 µA from the supply. Each tap is buffered by a TwoStageOpAmp instance configured as a unity-gain voltage follower (output tied to the inverting input), preventing the switching currents drawn by the MDAC sub-DAC capacitors and any transient loading from the comparator inputs from disturbing the ladder voltages.

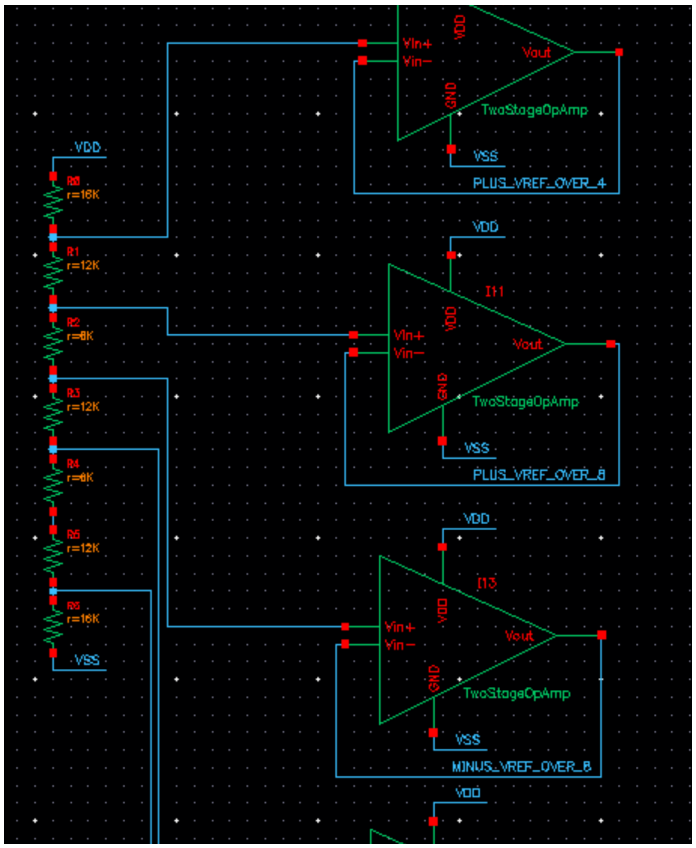


Figure 15: Reference voltage ladder and opamp buffers inside the `8b_pipeline_adc` schematic (close-up). The resistor ladder (R0–R6, left) divides V_{DD} to VSS into six taps. Three of the six unity-gain TwoStageOpAmp buffers are visible: PLUS_VREF_OVER_4 (3.25 V), PLUS_VREF_OVER_8 (2.875 V), and MINUS_VREF_OVER_8 (2.125 V).

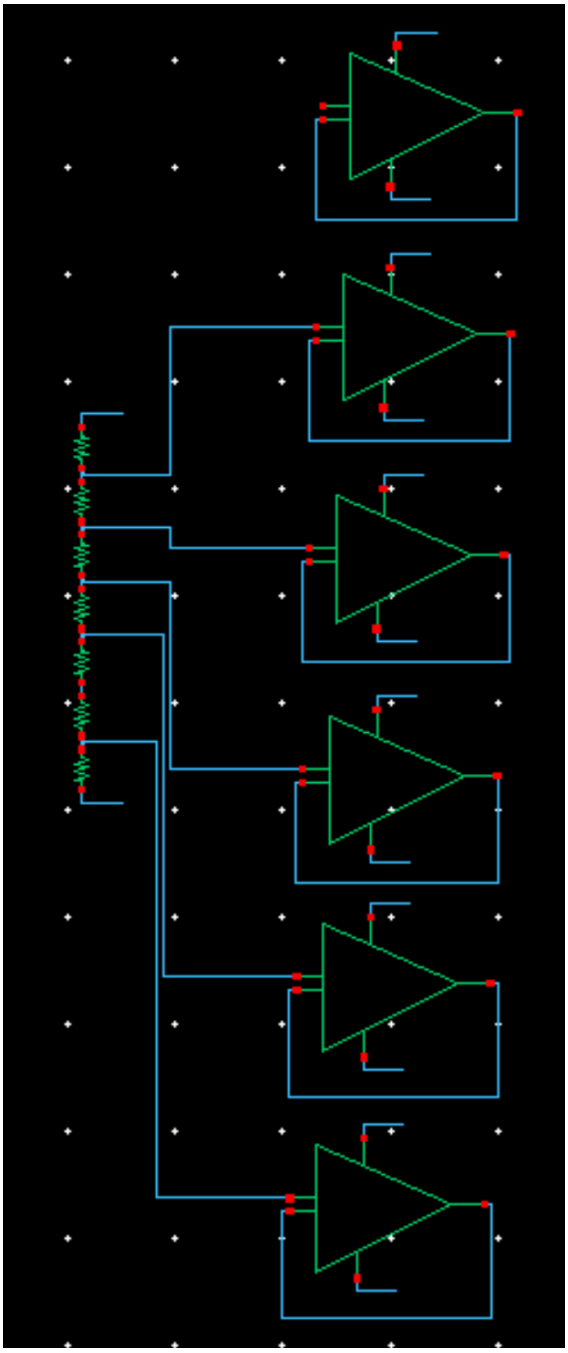


Figure 16: Full resistor ladder with all six unity-gain opamp buffers inside the `8b_pipeline_adc` schematic (zoomed out). Each ladder tap feeds a TwoStageOpAmp voltage follower, which drives the corresponding reference input to the 1.5-bit stages, flash back-end, or MDAC sub-DACs.

3.7 Digital Building Blocks

3.7.1 D Flip-Flop

A master-slave D flip-flop built from two NAND SR latches with clock-gated inputs and two inverters (10 NAND gates total, 44 transistors). All NAND gates use minimum-size transistors

($W/L = 1.5/0.6\,\mu$m for both NMOS and PMOS), and the two inverters are implemented as minimum-size CMOS inverters (`min_inv` cell, also $1.5/0.6\,\mu$m). Negative-edge triggered: the master latch is transparent when CLK is high, and the slave captures on the falling edge. 42 instances are used in the shift-register timing alignment network.

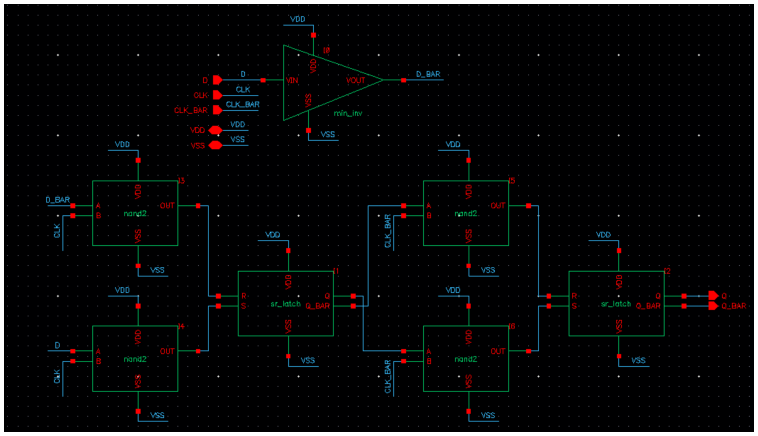
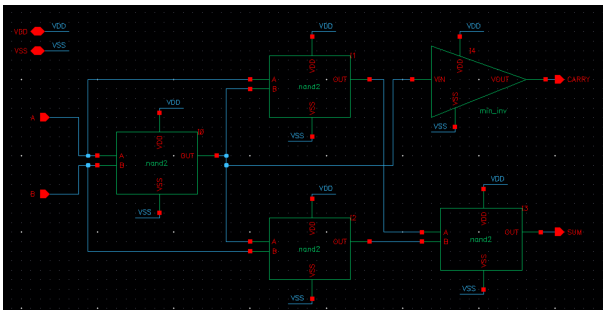


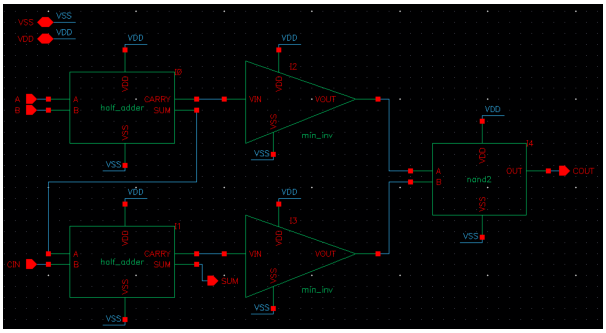
Figure 17: Master-slave D flip-flop from NAND gates.

3.7.2 Half Adder and Full Adder

The half adder uses 3 minimum-size NAND gates (`nand2`, $W/L = 1.5/0.6\,\mu$m for both NMOS and PMOS) and 2 minimum-size inverters (`min_inv`) to compute SUM $= A \oplus B$ and CARRY $= A \cdot B$. The full adder instantiates two half adders plus 2 `min_inv` inverters and 1 `nand2` gate (13 NAND-equivalent gates total), implementing the carry-out as COUT $=$ NAND$(\overline{C_1}, \overline{C_2}) = C_1 + C_2$ via De Morgan's law. All gates throughout the digital decoder are minimum-size; at the 2.5 MHz clock rate, propagation delays of a few nanoseconds through the NAND chains are negligible compared to the 200 ns clock half-period.



(a) Half adder (5 NAND gates): three NANDs and a min_inv compute SUM $= A \oplus B$, one NAND and a min_inv compute CARRY $= A \cdot B$.



(b) Full adder: two half_adder instances (I0, I1) plus two min_inv inverters and a NAND gate implement COUT $= C_1 + C_2$ via De Morgan's law. SUM comes from the second half adder directly.

Figure 18: Digital adder cells, both built entirely from nand2 and min_inv instances.

4 Inter-Stage Timing and Clock Phasing

Adjacent pipeline stages operate on alternating clock phases. This is implemented by feeding CLK_B to odd stages (1, 3, 5) and CLK_A to even stages (2, 4, 6) and the front-end SHA, with each stage's internal clock generator producing the correct $\phi_1/\phi_{1e}/\phi_2$ from whichever master clock it receives.

The SHA receives CLK_A, so it samples on CLK_A's ϕ_1 and holds on ϕ_2. Stage 1 receives CLK_B, so it samples the SHA's held output during CLK_B's ϕ_1. The flash back-end receives CLK_B, consistent with Stage 6 (even, CLK_A) amplifying on ϕ_2. No separate inter-stage sample-and-hold circuits are needed—each MDAC's input capacitors inherently perform this function.

The pipeline latency is 4 full clock periods (8 half-periods): 1 for the SHA + 6 stages + 1 for the flash, corresponding to 1.6 μs at $f_s = 2.5\,\mathrm{MHz}$.

5 Verification of Individual Components

5.1 MDAC Verification

The MDAC was tested standalone using the testbench shown in Figure 19. Ideal DC voltage sources provide $V_{\mathrm{DD}} = 5\,\mathrm{V}$, $V_{\mathrm{CM}} = 2.5\,\mathrm{V}$, PLUS_VREF = 4 V, and MINUS_VREF = 1 V. The bit inputs BI0 and BI1 are set to fixed DC values via design variables, and V_{in} is swept parametrically from 0 to 5 V in 20 steps. A vpulse source provides the clock, and V_{out} is sampled at the end of ϕ_2 (the amplify phase) to capture the settled residue. A 2 pF load capacitor on VOUT models the input capacitance of the next stage.

Three bit settings were tested, one for each 1.5-bit code:

- **High region** (BI0 = 5 V, BI1 = 5 V): $V_{\mathrm{out}} = 2V_{\mathrm{in}} - 4\,\mathrm{V}$. Figure 20a confirms slope 2 with the correct $-4\,\mathrm{V}$ offset—the markers read 212.5 mV at $V_{\mathrm{in}} = 2.1\,\mathrm{V}$ (ideal: $2 \times 2.1 - 4 = 200\,\mathrm{mV}$).
- **Mid region** (BI0 = 0 V, BI1 = 5 V): $V_{\mathrm{out}} = 2V_{\mathrm{in}} - 2.5\,\mathrm{V}$. Figure 20b shows 142.9 mV at $V_{\mathrm{in}} = 1.3\,\mathrm{V}$ (ideal: $2 \times 1.3 - 2.5 = 100\,\mathrm{mV}$).
- **Low region** (BI0 = 0 V, BI1 = 0 V): $V_{\mathrm{out}} = 2V_{\mathrm{in}} - 1\,\mathrm{V}$. Figure 20c shows 147.4 mV at $V_{\mathrm{in}} = 0.55\,\mathrm{V}$ (ideal: $2 \times 0.55 - 1 = 100\,\mathrm{mV}$).

All three curves exhibit the expected slope of 2 and clip at V_{DD} when the residue exceeds the opamp's output swing. Within the valid input range of 1–4 V, the residue error is within a few millivolts of ideal, confirming correct charge redistribution and sub-DAC switching for all three 1.5-bit codes.

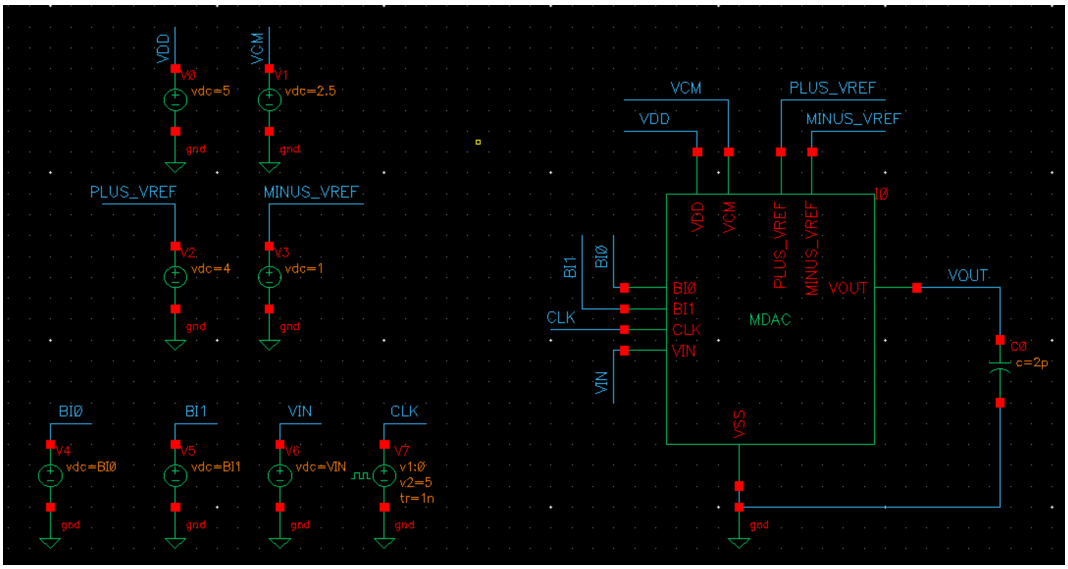
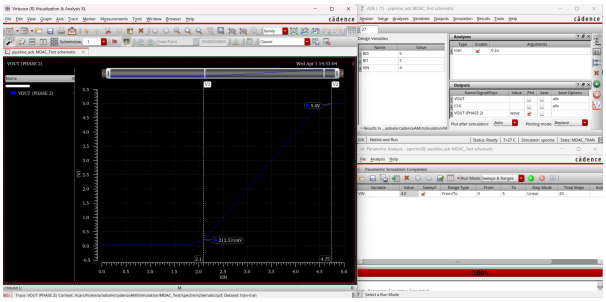
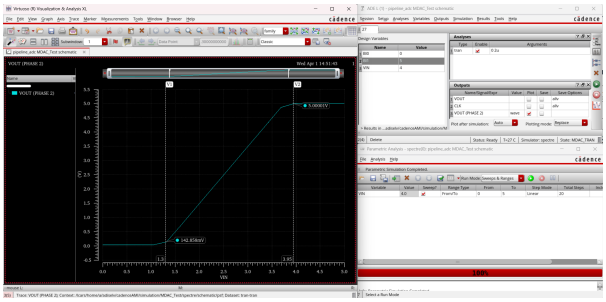


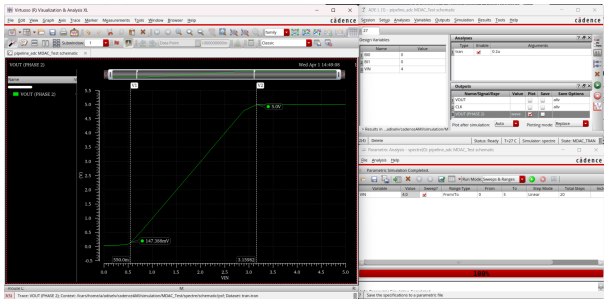
Figure 19: MDAC standalone testbench. Design variables BI0 and BI1 set the sub-DAC code; V_{in} is swept parametrically. VOUT is measured at the end of ϕ_2 with a 2 pF load capacitor.



(a) High region (BI0 = 5, BI1 = 5): $V_{\text{out}} = 2V_{\text{in}} - 4$.



(b) Mid region (BI0 = 0, BI1 = 5): $V_{\text{out}} = 2V_{\text{in}} - 2.5$.



(c) Low region (BI0 = 0, BI1 = 0): $V_{\text{out}} = 2V_{\text{in}} - 1$.

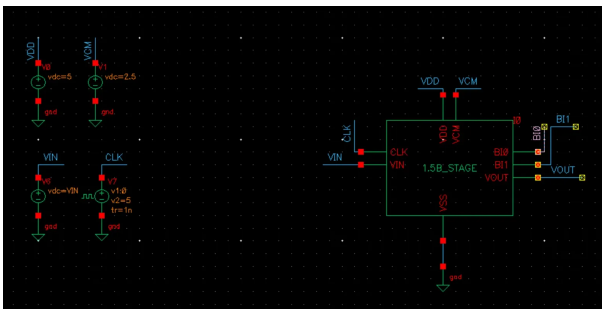
Figure 20: MDAC parametric sweep results for all three 1.5-bit codes. Each curve has slope 2 with the correct sub-DAC offset, confirming the charge-redistribution gain and reference switching.

5.2 Single 1.5-Bit Stage Verification

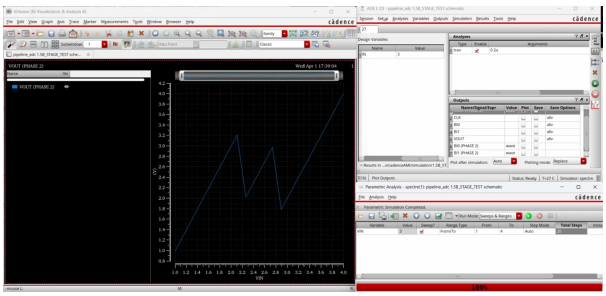
The complete 1.5-bit stage (with real StrongARM comparators and SR latches, not hardcoded bits) was tested using the testbench shown in Figure 21a. The setup is similar to the MDAC testbench, but now the entire `1.5B_STAGE` cell is instantiated—the comparators internally generate the bit

outputs BI0 and BI1 from the input signal. V_{in} is swept parametrically from 1 V to 4 V in 20 steps, and V_{out} is sampled at the end of ϕ_2.

Figure 21b shows the characteristic three-segment residue transfer function. The two transitions occur at $V_{\text{in}} \approx 2.125\,\text{V}$ and $\approx 2.875\,\text{V}$, confirming that the StrongARM comparators are switching at the correct thresholds. Each segment has slope 2, and the residue output remains bounded within approximately 1–4 V. The downward jumps at the transitions correspond to the sub-DAC switching from a lower to a higher reference voltage, subtracting a larger value and pulling the residue down. This plot confirms correct end-to-end operation of the sub-ADC, SR latches, and MDAC as an integrated unit.



(a) Single 1.5-bit stage testbench. $V_{\text{DD}} = 5\,\text{V}$, $V_{\text{CM}} = 2.5\,\text{V}$, V_{in} swept parametrically from 1 to 4 V. The `1.5B_STAGE` cell generates BI0, BI1, and VOUT internally.



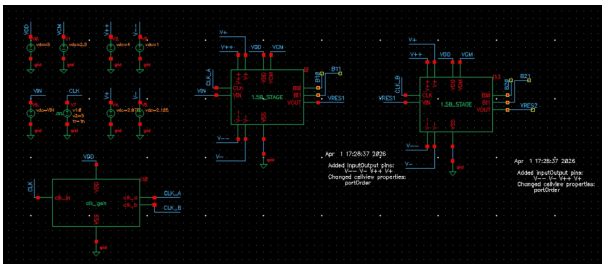
(b) Residue transfer function: three segments of slope 2 with transitions at 2.125 V and 2.875 V. Output bounded within ∼1–4 V, confirming correct comparator thresholds and sub-DAC switching.

Figure 21: Single 1.5-bit stage standalone verification.

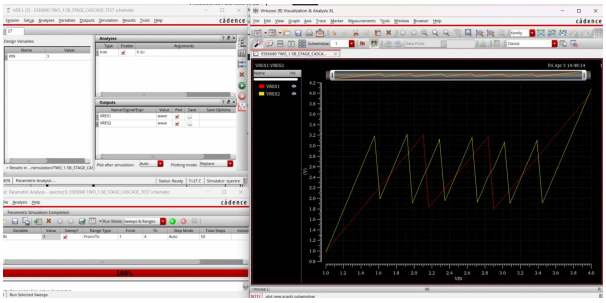
5.3 Two-Stage Cascade Verification

Two 1.5-bit stages were cascaded to verify inter-stage timing and residue amplification. The testbench (Figure 22a) uses a `clk_gen` instance to produce two non-overlapping clock phases CLK_A and CLK_B from a single master clock. Stage 1 receives CLK_A and Stage 2 receives CLK_B, so Stage 2 samples the residue from Stage 1 on the opposite phase. Ideal voltage sources provide the comparator thresholds ($V+ = 2.875\,\text{V}$, $V\text{-} = 2.125\,\text{V}$) and MDAC references ($V{+}{+} = 4\,\text{V}$, $V\text{-}\text{-} = 1\,\text{V}$). V_{in} is swept from 1 to 4 V in 50 steps, and both VRES1 (Stage 1 residue) and VRES2 (Stage 2 residue) are measured at the end of their respective amplify phases.

Figure 22b shows both residues overlaid. VRES1 (red) displays the familiar three-segment pattern with slope 2. VRES2 (yellow) shows the expected 9-segment "residue of residue" pattern (3×3 regions), where each of Stage 1's three segments is further subdivided by Stage 2's three comparator regions. Each VRES2 segment has slope 4 (2×2), confirming that the gain-of-2 from each stage compounds correctly through the cascade. The VRES2 output also remains bounded within approximately 1–4 V, verifying that the inter-stage signal range is preserved and that the alternating clock phasing produces correct sample-then-amplify sequencing between the two stages.



(a) Two-stage cascade testbench. A `clk_gen` produces CLK_A and CLK_B. Stage 1 receives CLK_A, Stage 2 receives CLK_B. VRES1 connects Stage 1 VOUT to Stage 2 VIN. V_{in} swept 1–4 V in 50 steps.



(b) Cascade residues: VRES1 (red, slope 2, 3 segments) and VRES2 (yellow, slope 4, 9 segments). The 3×3 subdivision and gain compounding confirm correct inter-stage operation.

Figure 22: Two-stage cascade verification with alternating clock phases.

5.4 2-Bit Flash ADC Verification

The 2-bit flash back-end was tested standalone using the testbench shown in Figure 23. Ideal DC sources provide $V_{\mathrm{DD}} = 5\,\mathrm{V}$, $V_{\mathrm{CM}} = 2.5\,\mathrm{V}$, $V+ = 3.25\,\mathrm{V}$, and $V- = 1.75\,\mathrm{V}$ as comparator reference thresholds. A vpulse provides the clock, and V_{in} is swept parametrically from 1 V to 4 V in 100 steps. The encoded outputs BI0 (b_{LSB}) and BI1 (b_{MSB}) are measured at the end of each clock period.

Figure 24 shows the results. BI1 (b_{MSB}, pink) transitions from low to high at $V_{\mathrm{in}} \approx 2.5\,\mathrm{V}$ and remains high, consistent with its direct connection to the middle comparator's SR latch. BI0 (b_{LSB}, blue) goes high at $\sim$1.75 V, returns low at $\sim$2.5 V, then goes high again at $\sim$3.25 V, producing the expected pattern for the thermometer-to-binary encoding $b_{\mathrm{LSB}} = C_1 \cdot \overline{C}_2 + C_3$. The four quantization regions (codes 00, 01, 10, 11) are clearly delineated at the correct threshold voltages.

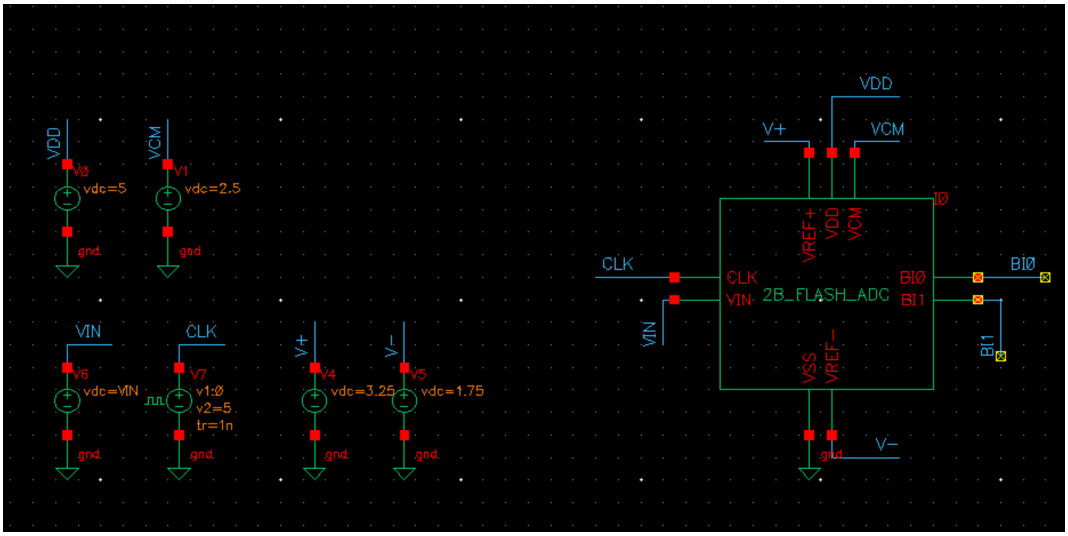


Figure 23: 2-bit flash ADC standalone testbench. Ideal voltage sources provide the reference thresholds ($V+ = 3.25\,\mathrm{V}$, $V- = 1.75\,\mathrm{V}$) and $V_{\mathrm{CM}} = 2.5\,\mathrm{V}$. V_{in} is swept parametrically from 1 to 4 V in 100 steps.

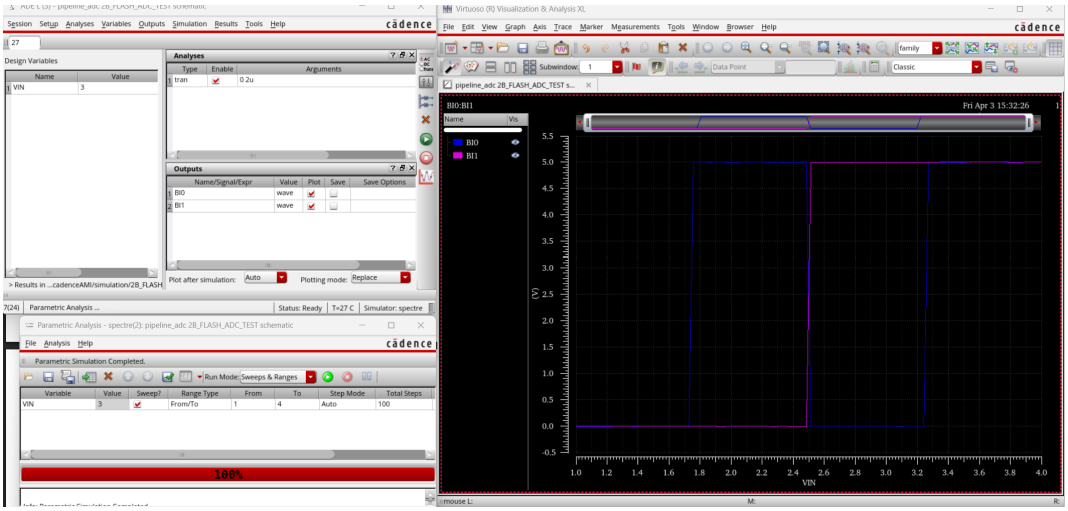


Figure 24: 2-bit flash ADC parametric sweep results. BI0 (blue, b_{LSB}) toggles at 1.75 V, 2.5 V, and 3.25 V. BI1 (pink, b_{MSB}) transitions once at 2.5 V. Together they produce the correct 4-level binary code: 00, 01, 10, 11.

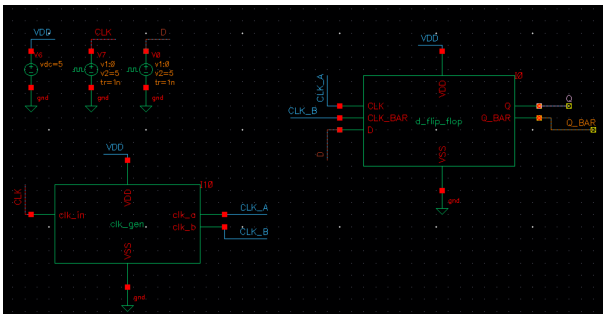
5.5 Digital Block Verification

Each digital building block was tested standalone with exhaustive input combinations before integration into the decoder.

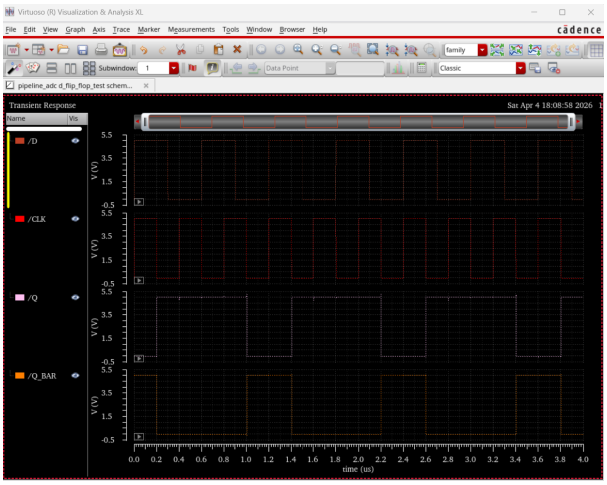
5.5.1 D Flip-Flop

The DFF testbench (Figure 25a) feeds a master clock through a `clk_gen` instance to produce CLK_A and CLK_B for the DFF's CLK and CLK_BAR inputs. The D input toggles at 1.5× the clock rate, so D changes mid-phase in some cycles, verifying that Q only updates on clock edges and ignores D transitions between them.

Figure 25b confirms correct negative-edge-triggered behavior: Q captures D on each falling edge of CLK and holds its value between edges regardless of changes on D. Q_BAR is the complement of Q at all times.



(a) DFF testbench. A `clk_gen` produces CLK_A and CLK_B from a vpulse master clock. D toggles at 1.5× the clock rate, ensuring D changes mid-phase to test the hold behavior.



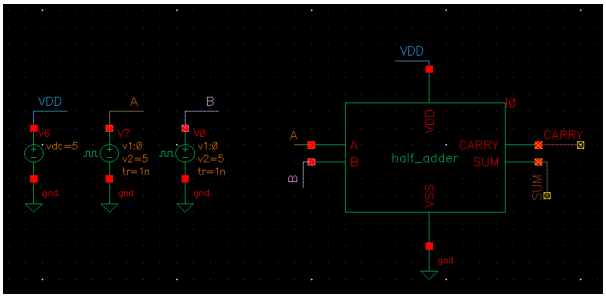
(b) DFF transient results. From bottom to top: D, CLK, Q, Q_BAR. Q updates only on CLK falling edges and holds between them; Q_BAR is always the complement of Q.

Figure 25: D flip-flop standalone verification.

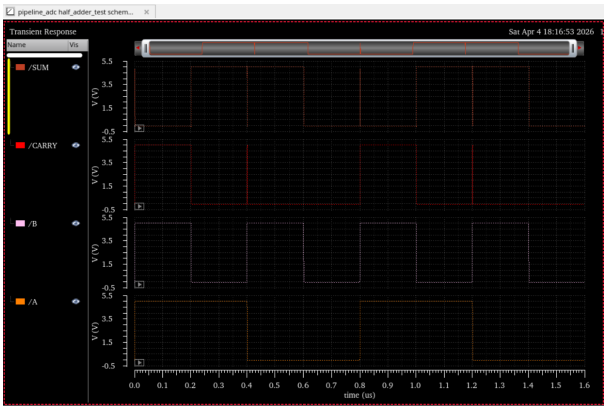
5.5.2 Half Adder

The half adder testbench (Figure 26a) uses two vpulse sources with periods of 800 ns (A) and 400 ns (B) to cycle through all four input combinations every 800 ns: $(0,0) \to (0,1) \to (1,0) \to (1,1)$.

Figure 26b shows the transient results over two full cycles. From bottom to top: A (orange), B (pink), CARRY (red), SUM (brick). SUM is high only when exactly one input is high, and CARRY is high only when both inputs are high, confirming correct XOR and AND functionality.



(a) Half adder testbench. Two vpulse sources (A at 800 ns period, B at 400 ns) cycle through all four input combinations.



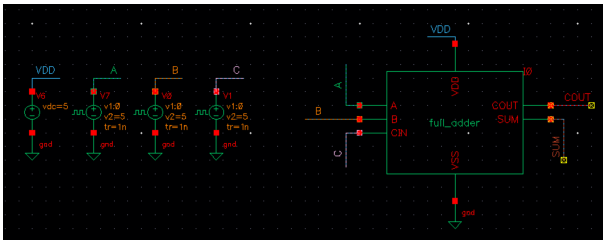
(b) Half adder transient results. Bottom to top: A, B, CARRY, SUM. CARRY goes high only for $(1,1)$; SUM goes high for $(0,1)$ and $(1,0)$.

Figure 26: Half adder standalone verification.

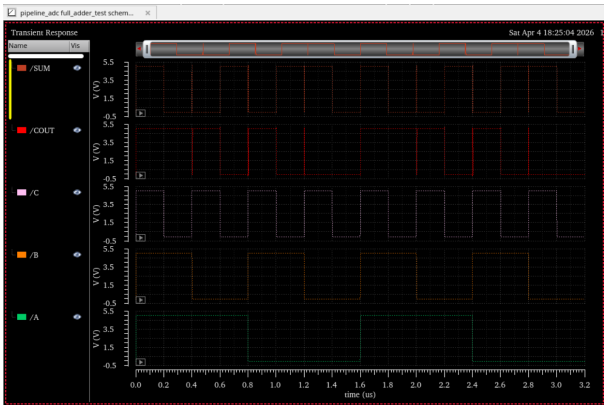
5.5.3 Full Adder

The full adder testbench (Figure 27a) uses three vpulse sources with periods of 1600 ns (A), 800 ns (B), and 400 ns (C) to cycle through all eight input combinations, one every 200 ns.

Figure 27b shows the transient results over two full cycles. From bottom to top: A (green), B (orange), C (pink), COUT (red), SUM (brick). SUM is high when an odd number of inputs are high; COUT is high when two or more inputs are high. Small glitch spikes are visible at transitions due to unequal propagation delays through the NAND chains, but all outputs settle to the correct values well within the 200 ns window.



(a) Full adder testbench. Three vpulse sources (A at 1600 ns, B at 800 ns, C at 400 ns) cycle through all eight input combinations.



(b) Full adder transient results. Bottom to top: A, B, C, COUT, SUM. Propagation delay glitches at transitions settle in nanoseconds.

Figure 27: Full adder standalone verification.

6 Full ADC Integration and Functional Verification

The complete 8-bit pipeline ADC was assembled as a single hierarchical cell (`8b_pipeline_adc`) with inputs CLK, VIN, VDD, VCM, and VSS, and eight digital outputs out_0 through out_7.

The testbench (Figure 28) applies a ramp input from 1 V to 4 V over 100 μs (vpulse V0 with $v_1 = 1$, $v_2 = 4$, $t_r = 100\,\mu$s), a vpulse clock, $V_{\mathrm{DD}} = 5$ V, and $V_{\mathrm{CM}} = 2.5$ V. An ideal 8-bit DAC (`dac_8bit_ideal` from `ahdlLib`, instance I1) converts the eight output bits back to an analog voltage for visual verification. The DAC's native output range is 0–1 V, so it is rescaled to the ADC's input range via DAC_rescaled = DAC_OUT $\times$ 3 + 1. The input ramp is time-shifted by $\Delta t = (4-1) \times (1.6/100) = 48$ mV equivalent, i.e. VIN_shifted = $V_{\mathrm{in}} - 0.048$ V, to compensate for the 1.6 μs pipeline latency (8 half-periods at $f_s = 2.5$ MHz). Overlaying the two traces allows accuracy to be judged at a glance.

Figure 29 shows the result. The rescaled DAC output (green staircase) tracks the latency-corrected input (cyan) closely across the full 1–4 V range. The initial transient at $t < 2\,\mu$s corresponds to the pipeline fill time before valid codes appear. This 100 μs ramp serves as a quick functional verification; the full static characterization (Section 7) uses a slower 500 μs ramp for more accurate DNL/INL measurement.

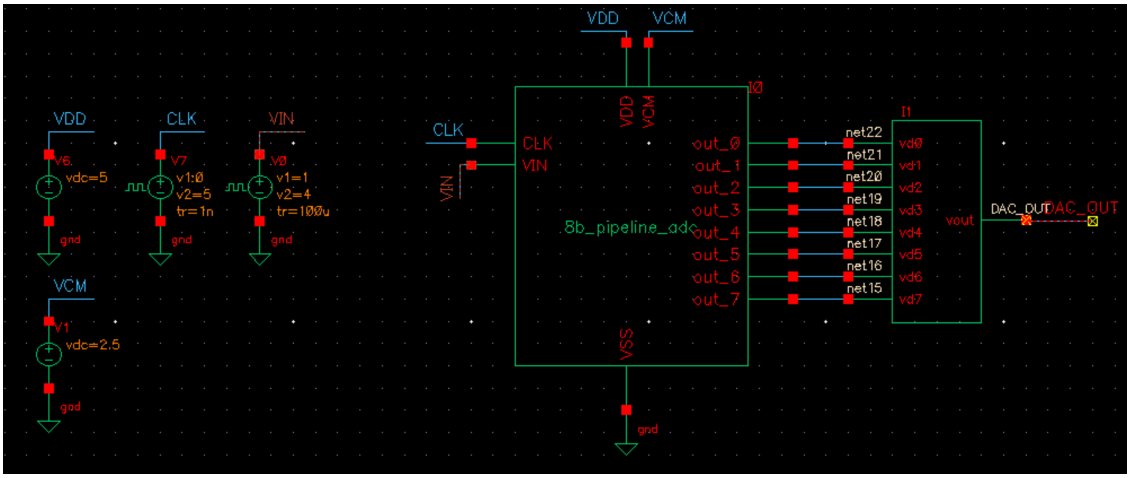


Figure 28: Full 8-bit pipeline ADC testbench. A 1–4 V ramp ($t_r = 100\,\mu$s) drives the `8b_pipeline_adc` cell (I0). The eight digital outputs connect to an ideal 8-bit DAC (I1) for reconstruction. $V_{DD} = 5\,$V, $V_{CM} = 2.5\,$V.

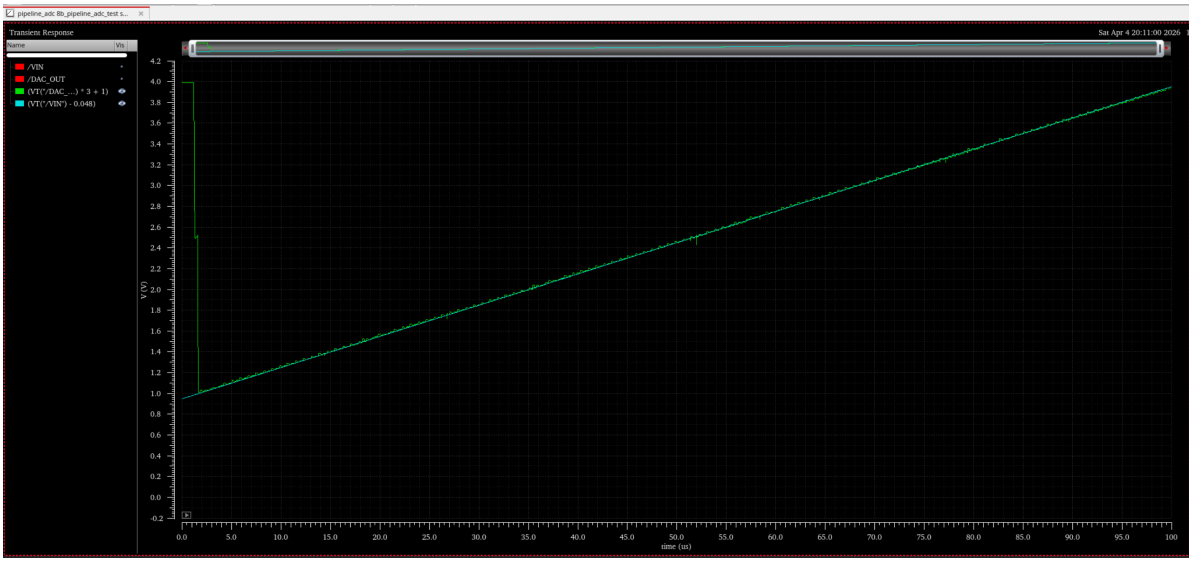


Figure 29: Full ADC ramp test. The rescaled DAC output (DAC_OUT $\times\, 3 + 1$, green) is overlaid with the latency-corrected input ($V_{in} - 0.048\,$V, cyan). The ADC tracks the input across the full 1–4 V range after the initial pipeline fill transient.

7 Static Performance Characterization

7.1 Test Setup

A slow linear ramp from 1 V to 4 V over 500 μs was applied to the full ADC testbench (same topology as Figure 28, but with $t_r = 500\,\mu$s). The total simulation length was 501.6 μs to allow the 1.6 μs pipeline to fill before valid codes appear. Both VIN and DAC_OUT were captured versus time and exported to CSV from Cadence's Visualization & Analysis XL.

7.2 Post-Processing Methodology

A Python script (run in Google Colab) processes the exported CSV to produce the transfer characteristic, DNL, and INL plots. The key steps are:

1. **Discard pipeline fill:** All samples before $t = 2\,\mu$s are removed to eliminate the startup transient where the pipeline has not yet filled with valid data.
2. **Code extraction:** The raw DAC output (0–1 V range) is converted to integer codes via code = round(DAC_OUT $\times$ 256), clipped to $[0, 255]$.
3. **Latency correction:** Since the ramp is linear, the pipeline latency ($\Delta t = 1.6\,\mu$s) maps to a fixed voltage offset. The corrected input is computed as $V_{\text{in,corr}} = 1 + 3 \times (t - 1.6\,\mu\text{s})/500\,\mu\text{s}$, aligning each output code with the input voltage that actually produced it.
4. **Transition detection:** For each pair of adjacent codes $k-1$ and k, the transition level $T(k)$ is taken as the midpoint between the last VIN sample that produced code $k-1$ and the first VIN sample that produced code k. This midpoint method averages out quantization noise at code boundaries.
5. **DNL computation:** The average code width is $W_{\text{avg}} = (T_{\text{last}} - T_{\text{first}})/(N_{\text{transitions}} - 1)$, serving as the measured LSB. DNL for code k is then $\text{DNL}(k) = (T(k+1) - T(k))/W_{\text{avg}} - 1$.
6. **INL computation:** INL is the deviation of each transition level from the best-fit line through the first and last transitions: $\text{INL}(k) = (T(k) - T_{\text{first}})/W_{\text{avg}} - (k - k_0)$.

7.3 Results

Figure 30 shows the transfer characteristic. The measured staircase (blue) tracks the ideal line (red dashed) closely across the full input range. The midscale zoom inset confirms uniform code widths around $V_{\text{cm}} = 2.5$ V with no missing codes, and the staircase steps are clearly resolved with no overlap or code skipping.

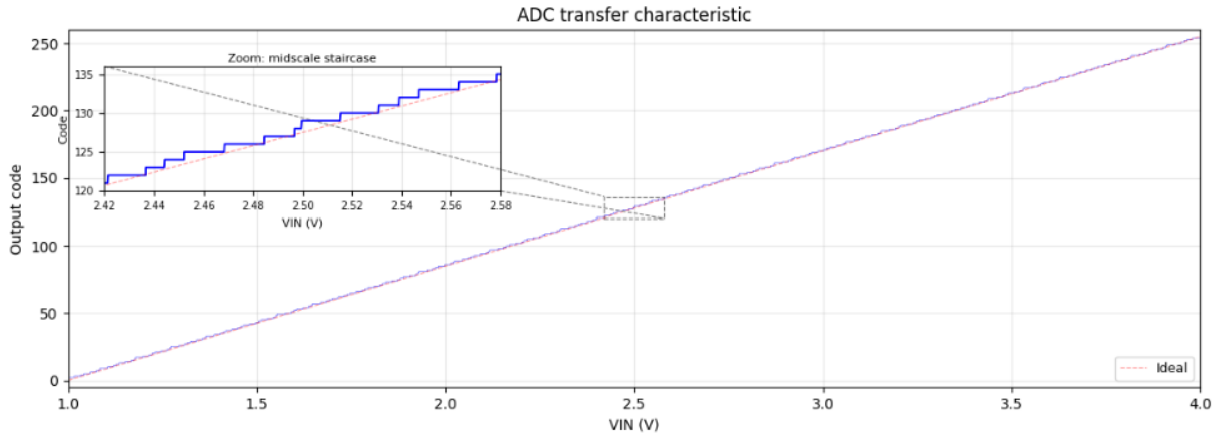


Figure 30: ADC transfer characteristic. Blue: measured output code vs. latency-corrected V_{in}. Red dashed: ideal linear mapping. Inset: zoom on the midscale region (2.42–2.58 V) showing individual code steps with no missing codes.

Figure 31 shows the DNL across all codes. The characteristic alternating ±DNL pattern—with positive DNL spikes near +0.9 LSB repeating every few codes—is a signature of systematic bit-weight error in the LSB positions. This pattern arises because finite opamp DC gain introduces a small but consistent residue error at each pipeline stage: the MDAC gain is not exactly 2 but $2 \times (1 - 1/(\beta A_v))$, so the effective bit weight of later stages is slightly compressed. Since this error

is deterministic and periodic (tied to the pipeline's binary structure), it manifests as a regular alternating DNL ripple rather than random code-width variation. Crucially, all DNL values remain within the ±1 LSB specification.

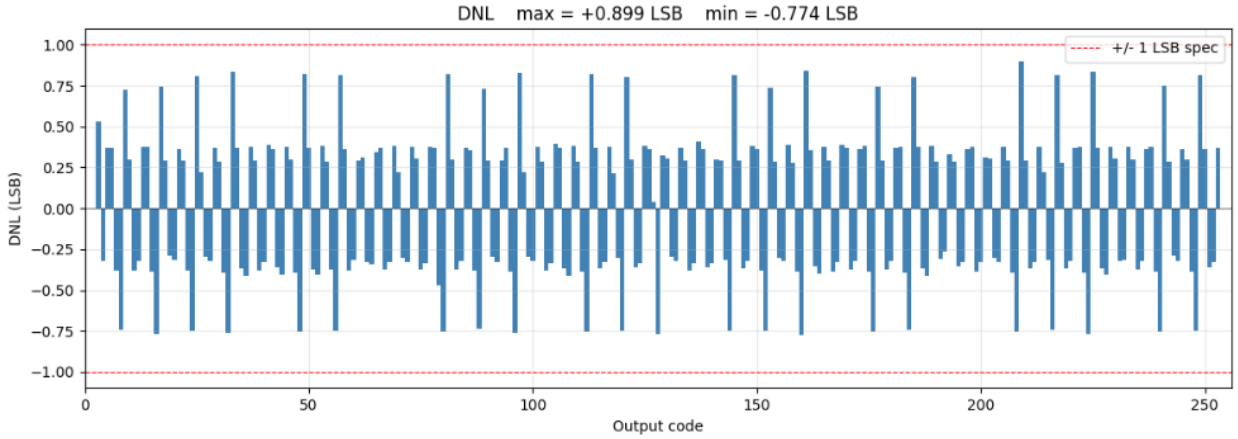


Figure 31: Differential nonlinearity (DNL). The alternating ± pattern across all codes is characteristic of systematic LSB bit-weight error from finite opamp gain. Max = +0.90 LSB, min = −0.77 LSB; both within the ±1 LSB specification (red dashed).

Figure 32 shows the INL, which exhibits a periodic oscillation with an envelope that is largest near the low codes and gradually decreases toward midscale and high codes. The oscillation period of approximately 4 codes corresponds to the Stage 6 bit cycling ($2^2 = 4$ codes per Stage 6 digit), suggesting that Stage 6's residue error is the dominant contributor—consistent with it being the last MDAC stage and therefore having the least settling margin relative to its significance. The decreasing envelope toward higher codes indicates that the gain error partially self-corrects as the input rises into a region where the opamp operates with better linearity (closer to V_{cm}). All INL values remain within ±1 LSB.

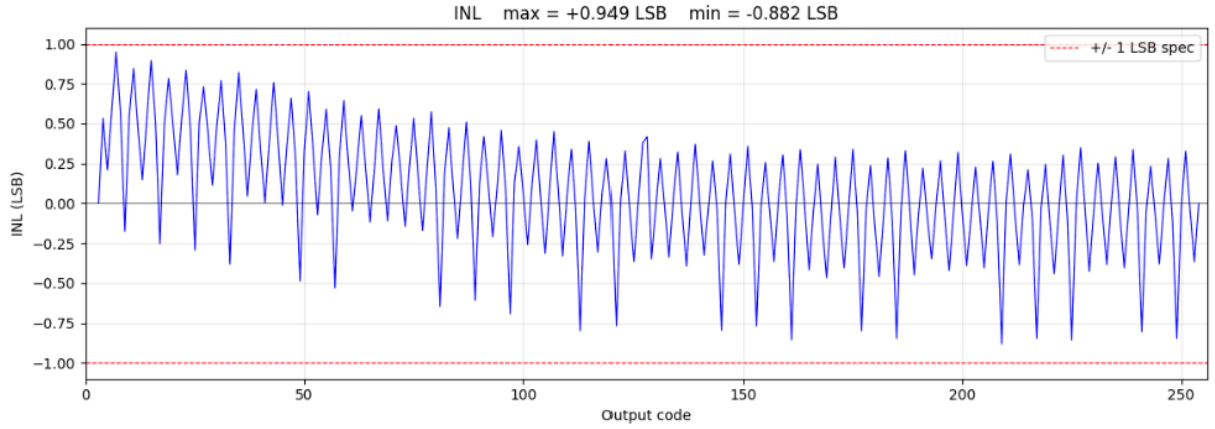


Figure 32: Integral nonlinearity (INL). The periodic oscillation with ∼4-code period tracks Stage 6 bit cycling. The envelope is largest near low codes and diminishes toward midscale. Max = +0.95 LSB, min = −0.88 LSB; both within the ±1 LSB specification (red dashed).

Table 7: Static performance summary.

Parameter	Spec	Measured	Status
Missing codes	0	0	PASS
DNL (max)	$\leq +1\,\text{LSB}$	$+0.90\,\text{LSB}$	PASS
DNL (min)	$\geq -1\,\text{LSB}$	$-0.77\,\text{LSB}$	PASS
INL (max)	$\leq +1\,\text{LSB}$	$+0.95\,\text{LSB}$	PASS
INL (min)	$\geq -1\,\text{LSB}$	$-0.88\,\text{LSB}$	PASS

The absence of missing codes and the margin to the $\pm 1\,\text{LSB}$ boundaries confirm that the 1.5-bit redundancy and digital error correction are functioning correctly: even though individual comparator offsets shift the residue transition points, the decoder recovers the true output code without introducing gaps or non-monotonicity.

8 Dynamic Performance Characterization

8.1 Test Setup

Dynamic performance was measured using coherent sampling to avoid spectral leakage. A full-scale sinusoidal input was applied with the following parameters:

$$f_s = 2.5\,\text{MHz}, \quad N = 256\,\text{samples}$$
$$M = 27\,(\text{signal bin, coprime with } N)$$
$$f_{\text{in}} = \frac{M}{N} \cdot f_s = \frac{27}{256} \times 2.5\,\text{MHz} = 263{,}672\,\text{Hz}$$

The input sine has $V_{\text{dc}} = 2.5\,\text{V}$ (common mode) and amplitude $1.5\,\text{V}$, sweeping the full 1–4 V input range. Since $M = 27$ and $N = 256$ share no common factors, each of the 256 samples falls on a unique phase of the sine cycle, guaranteeing that every quantization level is exercised and that no windowing is required. The same full ADC testbench from Section 6 was used, with the ramp source replaced by this sine. Both VIN and DAC_OUT versus time were exported to CSV from Cadence.

8.2 Post-Processing Methodology

A Python script (run in Google Colab) processes the exported CSV to produce the time-domain plot, FFT spectrum, and normalized spectrum. The key steps are:

1. **Sampling at clock edges:** The continuous-time DAC_OUT waveform is resampled at discrete clock edges by interpolating at $t_k = t_{\text{start}} + k \cdot T_{\text{clk}}$ for $k = 0, 1, \ldots$, where t_{start} accounts for the $1.6\,\mu\text{s}$ pipeline latency plus 350 ns to sample near the end of each clock period when the output has fully settled. The interpolated values are quantized to integer codes via code = round(DAC_OUT $\times$ 256).
2. **FFT:** The first $N = 256$ codes are transformed via a 256-point FFT. Magnitudes are normalized to the number of samples and referenced to the full-scale sinusoid amplitude (128 codes) to express the spectrum in dBFS.
3. **SNDR:** Computed as the ratio of signal power (bin $M = 27$) to total noise-plus-distortion power (all other non-DC bins): $\text{SNDR} = 10\log_{10}(P_{\text{signal}}/P_{\text{N+D}})$.
4. **ENOB:** Derived from SNDR via the standard relation $\text{ENOB} = (\text{SNDR} - 1.76)/6.02$.

5. **SFDR:** The ratio of the signal bin magnitude to the largest spurious bin magnitude (excluding DC): SFDR $= 20\log_{10}(|X[M]|/|X_{\text{spur,max}}|)$.
6. **Harmonics:** Bins corresponding to H_2 through H_5 are identified at $(h \cdot M)$ mod $(N/2)$ and their magnitudes are reported.

8.3 Results

Figure 33 shows the 256 sampled output codes in the time domain. The sinusoidal shape is clean, with codes spanning the full range from near 0 to near 255. The samples are evenly distributed across the sine cycle—a visual confirmation of the coherent sampling condition.

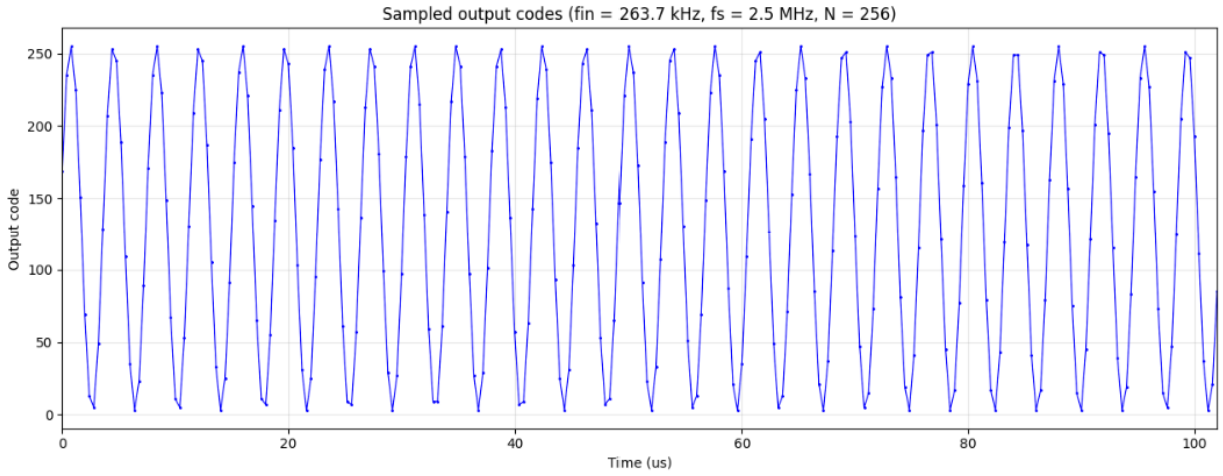


Figure 33: Time-domain sampled output codes. The full-scale sine ($f_{\text{in}} = 263.7\,\text{kHz}$, $f_s = 2.5\,\text{MHz}$) exercises the complete 0–255 code range. The 256 coherently sampled points are evenly distributed across the sine cycle.

Figure 34 shows the output spectrum in dBFS versus frequency. The signal at 263.7 kHz appears at -0.1 dBFS, confirming near-full-scale drive. The average noise floor sits at -67.5 dBFS. Harmonics H_2 through H_5 are visible above the noise floor, with the largest spur at bin 121 (1182 kHz, -52.3 dBFS), establishing an SFDR of 52.2 dB. This largest spur does not correspond to a simple integer harmonic of the input but rather to a folded intermodulation product—a consequence of the pipeline's nonlinear residue errors aliasing through the sampling process. The annotated double-arrow indicates the SFDR margin between the signal and the worst-case spur.

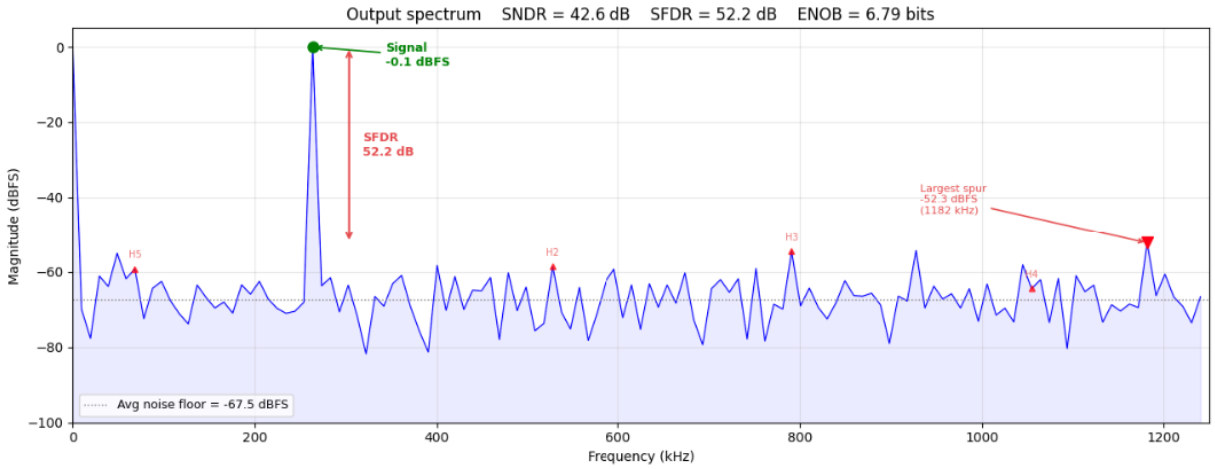


Figure 34: Output spectrum (dBFS vs. frequency). The signal at 263.7 kHz is at −0.1 dBFS. Harmonics H_2–H_5 are marked. The largest spur at 1182 kHz (−52.3 dBFS) sets the SFDR of 52.2 dB. Average noise floor is −67.5 dBFS (gray dotted).

Figure 35 shows the normalized spectrum with the noise-plus-distortion energy shaded in red. The signal bin stands clearly above the shaded region, and the SNDR of 42.6 dB and ENOB of 6.79 bits are annotated. For an ideal 8-bit ADC, the theoretical SNDR is $6.02 \times 8 + 1.76 = 49.9$ dB, corresponding to ENOB = 8.0. The measured ENOB of 6.79 represents a loss of approximately 1.2 bits.

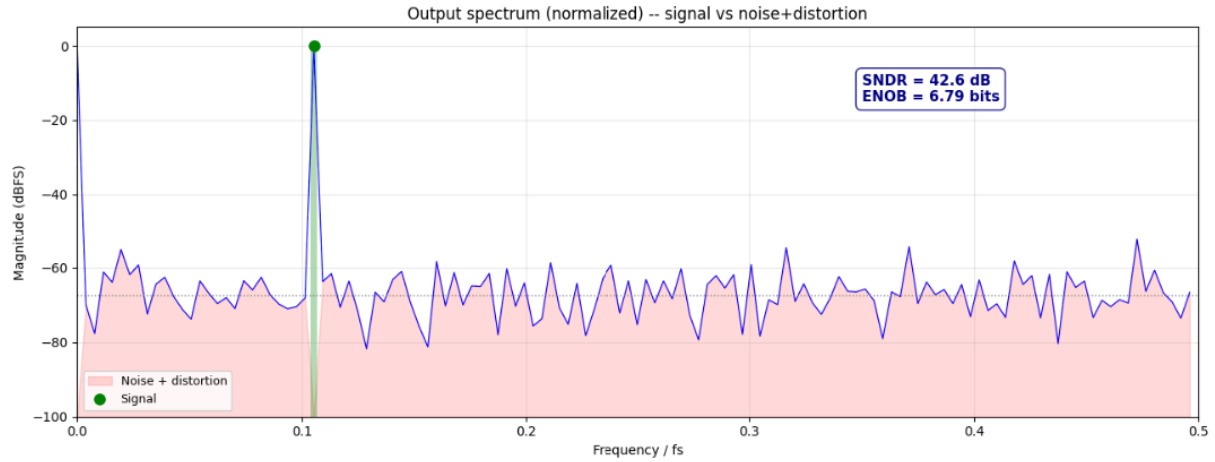


Figure 35: Normalized output spectrum (frequency axis in units of f_s). The signal bin (green dot) at $f/f_s \approx 0.105$ rises well above the noise-plus-distortion floor (red shading). SNDR = 42.6 dB, ENOB = 6.79 bits.

Table 8: Dynamic performance summary.

Parameter	Measured
f_{in}	263.7 kHz (bin 27)
f_s	2.5 MHz
N	256 samples
Signal magnitude	−0.1 dBFS
SNDR	42.6 dB
SFDR	52.2 dB
ENOB	6.79 bits
Noise floor (avg)	−67.5 dBFS
Largest spur	bin 121 (1182 kHz), −52.3 dBFS

8.4 Interpretation

The 1.2-bit ENOB loss relative to the ideal 8-bit ceiling is dominated by harmonic distortion rather than random noise, as evidenced by the discrete spurs rising above the noise floor. The primary mechanism is incomplete MDAC settling: with $f_{\text{CL}} = \beta \cdot \text{GBW} = 0.5 \times 27.2\,\text{MHz} = 13.6\,\text{MHz}$, the settling time constant is $\tau_{\text{CL}} = 1/(2\pi \times 13.6\,\text{MHz}) = 11.7\,\text{ns}$. The 200 ns amplification window provides $200/11.7 \approx 17$ time constants, which is generous for linear settling but marginal for the slew-limited portion: when the MDAC output must slew through the full $V_{\text{ref}}/2 = 1.5\,\text{V}$ step (at code transitions near $V_{\text{ref}}/8$ thresholds), the slew rate of $25.3\,\text{V}/\mu\text{s}$ requires 59 ns, consuming 30% of the settling window and leaving fewer linear time constants for final accuracy.

The SFDR of 52.2 dB exceeds the SNDR of 42.6 dB by nearly 10 dB, indicating that no single spur dominates the noise-plus-distortion budget; rather, the distortion energy is spread across multiple harmonic and intermodulation bins. This distributed distortion pattern is typical of pipeline ADCs where each stage contributes a small nonlinearity that compounds through the cascade.

Potential improvements include increasing the opamp GBW (via higher g_m or a folded-cascode topology) to provide more settling margin, or adopting a fully differential architecture to cancel even-order harmonics (H_2, H_4), which would improve SFDR and recover approximately 0.5 bits of ENOB.

9 Power Dissipation and Figure of Merit

9.1 Test Setup

Power was measured using the same testbench and sinusoidal input as the dynamic characterization ($f_{\text{in}} = 263.7\,\text{kHz}$, full-scale 1–4 V), but with the simulation length reduced to $10\,\mu\text{s}$ for tractable waveform export. Instead of capturing DAC_OUT, the supply current out of the V_{DD} source was recorded versus time and exported to CSV from Cadence.

9.2 Post-Processing Methodology

A Python script (run in Google Colab) processes the exported current waveform. The key steps are:

1. **Discard startup:** All samples before $t = 2\,\mu\text{s}$ are excluded to eliminate the pipeline fill transient, where the current waveform has not yet reached steady-state periodicity.

2. **Average power:** The mean supply current over the steady-state region is computed via trapezoidal integration: $I_{\text{avg}} = \frac{1}{t_{\text{end}} - t_{\text{start}}} \int_{t_{\text{start}}}^{t_{\text{end}}} i(t)\, dt$, and average power follows as $P_{\text{avg}} = V_{\text{DD}} \times I_{\text{avg}}$.
3. **Per-cycle power:** The simulation is partitioned into individual 400 ns clock cycles ($T_{\text{clk}} = 1/f_s$). For each cycle, the average current is integrated separately, yielding a per-cycle power value that reveals how power consumption varies with the instantaneous input signal level.
4. **Worst-case identification:** The clock cycle with the highest average power is identified, and its timing is mapped back to the input sine phase to determine under what signal conditions the ADC draws maximum current.

9.3 Results

Figure 36 presents the complete power analysis across six sub-panels.

Panel (a) shows the raw V_{DD} supply current over the full 10 μs simulation. The gray-shaded startup region ($t < 2\,\mu$s) is excluded from all calculations. The steady-state current oscillates between a baseline of $\sim$2.5 mA (the static bias of the seven opamps) and sharp spikes up to $\sim$14 mA during clock transitions, when all switched-capacitor circuits and digital logic switch simultaneously. The average current (red dashed) is 3.585 mA.

Panel (b) shows the average power per clock cycle. Each bar represents the energy consumed in one 400 ns period. The worst-case cycle (red bar) draws 18.50 mW, while the average across all steady-state cycles is 17.92 mW. The cycle-to-cycle variation of $\sim$15% reflects the signal-dependent dynamic switching current.

Panel (c) overlays the input sine wave with the worst-case cycle highlighted. The worst cycle occurs at $t = 4.2\,\mu$s, corresponding to a sine phase of approximately 39° and $V_{\text{in}} = 3.44$ V. This is near the steepest portion of the sine (dV/dt is maximum near the zero crossing in terms of AC swing), where the MDAC capacitors must slew through the largest inter-sample voltage steps, producing maximum dynamic switching current.

Panel (d) zooms in on the supply current around the worst-case cycle. The current spikes correlate with clock edges: each rising and falling edge triggers simultaneous switching of transmission gates across all six MDAC stages and the SHA, producing transient current pulses that decay as the capacitors settle.

Panel (e) shows the power distribution histogram across all steady-state cycles. The distribution spans 16.0–18.5 mW with most cycles clustered around 17.0–17.5 mW, confirming that the worst-case is only marginally above the average.

Panel (f) summarizes the FOM computation.

$$P_{\text{avg}} = V_{\text{DD}} \times I_{\text{avg}} = 5\,\text{V} \times 3.585\,\text{mA} = 17.92\,\text{mW} \tag{7}$$

The static power is dominated by the seven opamp instances (6 MDAC stages + 1 SHA), each biased at $I_{\text{ref}} = 20\,\mu$A with a total quiescent current of $\sim$0.17 mA per opamp, contributing $\sim$0.85 mW each for a total of $\sim$5.95 mW. The remaining $\sim$12 mW is dynamic power from switched-capacitor charge transfer, clock driver switching, digital logic transitions, and the six opamp buffers in the reference voltage ladder.

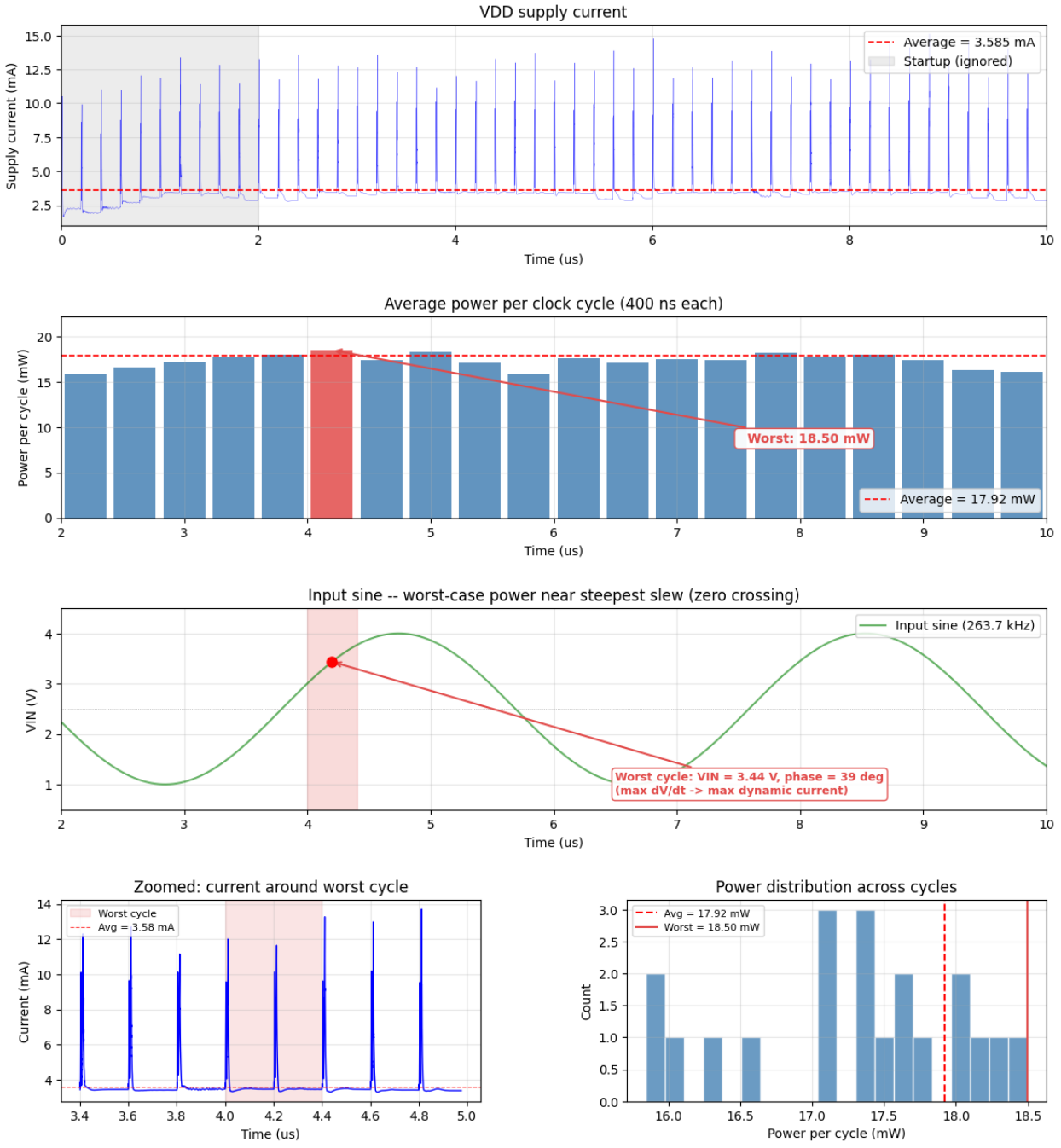


Figure 36: Power analysis. (a) Raw V_{DD} supply current over 10 μs; gray region is startup (excluded). Average = 3.585 mA. (b) Per-cycle average power (400 ns each); worst cycle (red) = 18.50 mW, average (dashed) = 17.92 mW. (c) Input sine with worst-case cycle highlighted at $t = 4.2\,\mu$s (phase $\approx 39°$, near maximum dV/dt). (d) Zoomed current around worst cycle showing switching transients. (e) Power histogram across all steady-state cycles. (f) FOM calculation summary.

9.4 Figure of Merit

The Walden figure of merit is defined as:

$$\text{FOM} = \frac{P_{\text{avg}}}{f_s \times 2^B} \tag{8}$$

where B is either the nominal resolution or the measured ENOB. Using the nominal 8-bit resolution gives an optimistic FOM, while using the measured ENOB of 6.79 bits gives the effective FOM that accounts for the actual conversion quality.

Table 9: FOM summary using average and worst-case power.

	Power	$\mathbf{2^B}$	**FOM (pJ/conv-step)**
Average power (17.92 mW):			
Ideal ($B = 8$)	17.92 mW	256	28.0
Actual (ENOB = 6.79)	17.92 mW	110.5	**64.8**
Worst-case power (18.50 mW):			
Ideal ($B = 8$)	18.50 mW	256	28.9
Actual (ENOB = 6.79)	18.50 mW	110.5	66.9

The effective FOM of 64.8 pJ/conv-step is dominated by the opamp power. The single-ended architecture requires each opamp to bias at a current sufficient for the full output swing (1–4 V), and the seven instances account for the majority of the static dissipation. A fully differential design would improve both ENOB (by canceling even-order harmonics) and power efficiency (by halving the required output swing per side), directly reducing the FOM on both axes.

10 Maximum Sampling Rate Estimation

The maximum achievable sampling rate $f_{s,\max}$ is limited by the time required for the MDAC to settle to 8-bit accuracy during the amplify phase. The amplify phase duration is $T_{\phi_2} = 1/(2f_s)$, and must accommodate both the slew-limited and exponentially-settling portions of the MDAC output transient.

10.1 MDAC Settling Budget

The MDAC settling transient consists of two sequential phases:

1. Slew-limited phase. When the MDAC output must traverse a large voltage step (e.g., at a code boundary where the sub-DAC reference switches), the opamp slews at its maximum rate until the output enters the linear settling region. The worst-case output step occurs at the $V_{\text{ref}}/8$ thresholds, where the sub-DAC switches between $+V_{\text{ref}}/2 = 4\,\text{V}$ and $-V_{\text{ref}}/2 = 1\,\text{V}$, producing a residue jump of $\Delta V_{\max} = V_{\text{ref}}/2 = 1.5\,\text{V}$. The time spent slewing is:

$$t_{\text{slew}} = \frac{\Delta V_{\max}}{\text{SR}} = \frac{1.5\,\text{V}}{25.3\,\text{V}/\mu\text{s}} = 59.3\,\text{ns} \tag{9}$$

2. Exponential settling phase. After the slew completes, the remaining error decays exponentially with the closed-loop time constant τ_{CL}:

$$\tau_{\text{CL}} = \frac{1}{2\pi \cdot \beta \cdot \text{GBW}} = \frac{1}{2\pi \times 0.5 \times 27.2\,\text{MHz}} = 11.7\,\text{ns} \tag{10}$$

For 8-bit accuracy, the residue must settle to within $\frac{1}{2}$ LSB of the 9-bit equivalent (since first-stage errors are not attenuated). The required number of time constants is:

$$n = \ln(2^{N+1}) = \ln(512) = 6.24 \tag{11}$$

giving a linear settling time of $t_{\text{linear}} = 6.24 \times 11.7\,\text{ns} = 73.0\,\text{ns}$.

However, this assumes a purely exponential step response. With a phase margin of 70.9°, the response is well-damped but not perfectly first-order. A more conservative estimate accounts for the second-order ringing by using $n \approx 9$ time constants:

$$t_{\text{linear,conservative}} = 9 \times 11.7\,\text{ns} = 105.3\,\text{ns} \tag{12}$$

10.2 Maximum Sampling Rate

The total amplify phase duration must satisfy $T_{\phi_2} \geq t_{\text{slew}} + t_{\text{linear}}$:

Table 10: Maximum sampling rate estimates from measured opamp parameters.

Estimate	$\boldsymbol{T_{\phi_2,\text{min}}}$	$\boldsymbol{f_{s,\text{max}}}$	**Margin at 2.5 MHz**
Moderate ($n = 6.24\,\tau$)	132.3 ns	3.78 MHz	1.51×
Conservative ($n = 9\,\tau$)	164.6 ns	3.04 MHz	1.22×

At the design sampling rate of $f_s = 2.5\,\text{MHz}$ ($T_{\phi_2} = 200\,\text{ns}$), the settling budget has a margin factor of 1.22–1.51× beyond what is strictly required, consistent with the measured ENOB of 6.79 bits (close to but not fully achieving 8-bit accuracy).

10.3 Other Timing Constraints

Two additional timing constraints were verified to confirm they do not limit $f_{s,\text{max}}$:

Signal acquisition. The sampling phase must allow the TG to charge the 2 pF capacitor to within $\frac{1}{2}$ LSB. With worst-case $R_{\text{on}} \approx 1.3\,\text{k}\Omega$, the acquisition time constant is $\tau_{\text{acq}} = 2.6\,\text{ns}$. For 9-bit settling accuracy, $t_{\text{acq}} = 6.24 \times 2.6 = 16.2\,\text{ns} \ll 200\,\text{ns}$. Signal acquisition does not limit f_s.

Comparator decision. The StrongARM comparator resolves a 50 mV differential in 1.15 ns, and the SR latch settles within a few additional nanoseconds. The total sub-ADC decision time of $< 5\,\text{ns}$ is negligible compared to the amplify phase and does not constrain f_s.

10.4 Improving $f_{s,\text{max}}$

The settling budget is dominated by the slew-limited phase (59.3 ns out of 132–165 ns total), suggesting that increasing the opamp slew rate would be the most effective way to raise $f_{s,\text{max}}$. Since $\text{SR} = I_{\text{tail}}/C_c$, doubling the bias current (with proportional C_c adjustment to maintain phase margin) would roughly halve t_{slew} to ∼30 ns, pushing $f_{s,\text{max}}$ to ∼5 MHz. Alternatively, a folded-cascode opamp topology would provide higher GBW at the same power, reducing τ_{CL} and compressing the linear settling time.

11 Performance Summary

Table 11: Complete ADC performance summary.

Parameter	Specification	Measured
Supply	5 V	5 V
f_s	> 2 MHz	2.5 MHz
$f_{s,\max}$ (estimated)	—	3.0–3.8 MHz
Input range	1–4 V	1–4 V
Resolution	8 bits	8 bits
DNL	≤ 1 LSB	$+0.90/-0.77$ LSB
INL	≤ 1 LSB	$+0.95/-0.88$ LSB
SNDR	—	42.6 dB
SFDR	—	52.2 dB
ENOB	—	6.79 bits
Avg power	Minimize	17.92 mW
FOM (ENOB)	Minimize	64.8 pJ/conv-step

12 Contributions

Table 12: Team member contributions.

Member	Contributions
Adithya Selvakumar	MDAC switched-capacitor design and sub-DAC mux logic, 1.5-bit stage integration and debugging (SR latch polarity fix, comparator output routing), 2-bit flash thermometer-to-binary encoder, digital decoder architecture (DFF, half/full adders, shift-register timing alignment), full ADC integration and top-level wiring
Krishna Chemudupati	Two-stage Miller-compensated opamp design and characterization (Project 1), StrongARM dynamic latch comparator design (Project 2), flip-around SHA design and verification (Project 2), non-overlapping clock generator design, CMOS transmission gate design, NAND SR latch design, reference voltage ladder with unity-gain opamp buffers, stage cascading and inter-stage clock phasing verification
Shared	All post-processing scripts (DNL/INL, SNDR/SFDR/ENOB, power/FOM), report writing

13 Conclusion

A fully functional 8-bit pipeline ADC was designed and characterized in the AMI C5N 0.6 μm process. The converter meets all static specifications: DNL of $+0.90/-0.77$ LSB and INL of $+0.95/-0.88$ LSB, both within the ± 1 LSB requirement, with no missing codes across the full

1–4 V input range. Dynamic testing at 2.5 MSa/s yields an SNDR of 42.6 dB, SFDR of 52.2 dB, and ENOB of 6.79 bits, with average power consumption of 17.92 mW (FOM = 64.8 pJ/conv-step using ENOB).

All circuit blocks—including the opamp, comparators, SR latches, clock generators, transmission gates, SHA, MDAC, flash ADC, D flip-flops, half adders, and full adders—were built from scratch as custom transistor-level schematics in Cadence. No ideal components appear in the final ADC; only the characterization DAC used for reconstruction during testing is an ideal library element. The design reuses the opamp (Project 1), comparator, SHA, clock generator, and transmission gate (Project 2), demonstrating a complete bottom-up mixed-signal design flow from individual transistors to a system-level converter.

The 1.2-bit gap between the nominal 8-bit resolution and the measured ENOB is attributable primarily to harmonic distortion from two sources: finite opamp DC gain causing systematic bit-weight compression in later pipeline stages (visible as the periodic DNL ripple), and incomplete MDAC settling during the slew-limited portion of large residue transitions. With GBW = 27.2 MHz and $\beta = 0.5$, the closed-loop bandwidth of 13.6 MHz provides adequate linear settling margin but leaves limited headroom after worst-case slewing consumes 30% of the amplification window.

Several avenues exist for improving performance without fundamentally changing the architecture. Increasing the opamp g_m (via larger input pair W/L or higher bias current) would widen the GBW and reduce settling error. A folded-cascode first stage would provide higher DC gain in a single stage, reducing the systematic bit-weight error that dominates DNL. Most significantly, a fully differential signal path would cancel even-order harmonics (H_2, H_4), potentially recovering 0.5–1 bits of ENOB while also improving power supply rejection and common-mode noise immunity.